This Step or From Here On? Crossed Same-State Interventions for Evaluating Quantized Agents

Anonymous ACL submission

Abstract

When a quantized agent and its BF16 reference act differently at a state, should BF16 supply this step, take over from the next step, or both? We measure this with a crossed same-state intervention: at a frozen state that replays bit-exactly, a 2×2 factorial design crosses the source of the current action (BF16 or quantized) with the controller of the rest of the episode. We study Qwen3-4B and Qwen3-8B under simulated weight and KV-cache quantization with BF16 compute, using one greedy user simulator in τ^2-bench and forking at the first decision where the two models' actions differ in tool structure. We report three findings. (i) The value of an action depends on the controller that continues. In 26.9% of development W4 divergent states, the same action swap has a different outcome effect under BF16 and under quantized continuation, and this dependence also appears, to differing degrees, on BFCL and ALFWorld, which have no LLM user. (ii) Local upgrades redistribute success and failure. Replacing a W4 agent's action rescues or damages 15.5–20.9% of task-block units, with a near-zero net effect on development tasks. On 34 pre-registered unseen telecom tasks, the swap's estimated effect was -4.6 pp (90% CI -7.2 to -2.0); the pre-registered readings, which had no negative category, classify it as inconclusive. A same-precision baseline, with a second GPTQ-W4 calibration draw in place of BF16, tests whether turnover and controller dependence are specific to precision: [same-precision reading pending]. (iii) A frozen uncertainty trigger finds outcome-changing decisions more often than random selection at two of three rates, but its net gain is not significant after correction. Precision upgrades for agents should be judged by the task value of the replacement under the execution that follows it.

1 Introduction

Quantized LLM agents are usually compared with their full-precision reference by average task success. In closed-loop benchmarks such as τ^2-bench (Barres et al., 2025), that comparison cannot distinguish difference from equivalence. On our checkpoints, quantized-minus-BF16 success ranges from -0.2 to $+4.2$ pp across four W4 cells, and the new-task (crossed) intervals are about ± 6–7 pp wide, so an equivalence test at ± 3 pp fails in every cell (Appendix B). Yet the quantized agent (Q) still takes actions that its BF16 reference (B) would not. If B is available as a fallback, as in dynamic precision routing (Li et al., 2026c), each such state poses a decision: **upgrade precision for this step only, from the next step on, or both?** The answer depends on the action the upgrade changes and on the execution that follows it.

Consider one state from our data (q4 AWQ-W4, τ^2-bench airline task 31, decision 1 of 12). Q looks up a reservation id that does not exist and receives "not found"; B looks up the user's profile. B's action looks better. We froze this state and ran all four combinations of {B's or Q's action} $\times$ {B or Q continuing}. Only B's action followed by B succeeds. If B supplies only this step, Q continues, searches for flights, writes a change and fails. If Q's failed lookup stands and B takes over, B fails too. B's action helps only if B also continues, which no judgement of the action alone can reveal.

A crossed same-state intervention. At a frozen state whose no-op replay is bit-exact, we cross the source of the current action with the controller of the remaining episode (Figure 1). This turns "this step" (G), "from the next step" (H) and "both" (F) into directly comparable interventions on the same states (§3). We study Qwen3-4B and Qwen3-8B (simulated weight and KV-cache quantization, BF16 compute) in τ^2-bench with one greedy user

simulator, and in BFCL multi-turn and ALFWorld. We fork at the first decision where the two models' actions differ in tool structure (text-only differences are skipped) and, separately, at ordinary states drawn independently of any outcome.

What we find.

- **The value of an action depends on the controller that continues** (§5). In 26.9% of development W4 divergent states, the same action swap has a different outcome effect under B's continuation than under Q's; strict reversals are rare. A post hoc analysis finds the same dependence on BFCL (4.7–17.1% per W4 cell) and q8 ALFWorld (32.8–40.0%), which have no LLM user, and on the telecom holdout (20.7%). Cross-cell comparisons are descriptive, and we establish no $G-H$ ordering that transfers across cells.
- **Local precision upgrades redistribute success and failure** (§4, §6). Replacing a W4 agent's first tool-structure-divergent action rescues or damages 15.5–20.9% of task×block units, with a net effect of -1.9 to $+2.9$ pp on development tasks. On 34 pre-registered unseen telecom tasks, the swap again rescues and damages and task fragility repeats; its estimated effect was -4.6 pp (90% CI -7.2 to -2.0), which the pre-registered readings, having no negative category, classify as inconclusive. A same-precision baseline (a second GPTQ-W4 calibration draw in place of B) tests whether this turnover and the controller dependence above are specific to precision: [same-precision reading pending].
- **Finding sensitive decisions and choosing beneficial replacements are different targets** (§7). A frozen Q-uncertainty trigger captures more outcome-changing decisions than random selection at two of three rates on later nonce blocks, but its net gain over random selection is not significant after correction.

Contributions. (1) A crossed-intervention framework for evaluating precision upgrades in agents. Forking and single-action replay exist (§2); we add the crossing of the action with the continuation controller (G, H, F and the (Δ_Q, Δ_B) table), bit-exact replay acceptance, explicit denominators, and inference that separates a fixed task pool from new tasks. (2) A study over weight, KV-cache and stress quantization, two model sizes and three benchmarks, covering two-sided consequences, controller dependence across environments, text actions at ordinary states (whose outcome labels did not reproduce under a second simulator; §6) and diagnostics for precision selection. Negative and withdrawn results are in Appendix L.

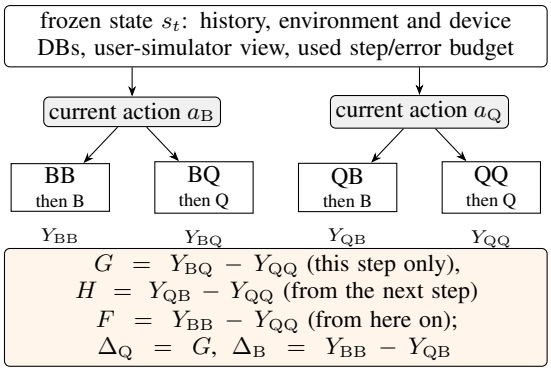


Figure 1: Crossed same-state intervention (schematic, not data). B = BF16 agent, Q = quantized agent. In branch XY, the first letter is the source of the current action and the second is the controller from the next step to the end. QQ re-executes Q's original episode, and its bit-exact reproduction is the no-op check. Definitions are in §3.2.

2 Related Work

Episode-level comparisons of quantized agents. *Flat Score, Amplified Failures* (Jang et al., 2026) is the closest study in setting. On τ^2-bench retail and telecom it shows that 4-bit agents keep a flat task score while process damage grows. It reports a two-sided paired task split (33 tasks worse, 26 better for Gemma-4-31B on telecom) and explains the flat score by the error budget and by recovery, equally frequent at both precisions (55% after a failed call). Its comparisons are between independently sampled episodes, so the precision of the action that went wrong and the precision of the controller that must recover change together. Our design holds the state fixed and varies the two separately. H measures what the recovering controller contributes on identical histories, and the (Δ_Q, Δ_B) table measures whether an action's consequence depends on that controller. Two-sided flips under compression were first shown on static tasks (Dutta et al., 2024); agent-compression benchmarks report averages (Dong et al., 2025).

Forks and counterfactual replay. *The Replay Gap* (Gonuguntla, 2026) forks live SWE-agent trajectories at a fixed step and continues them with a different model (Qwen3-4B FP8 vs. Qwen3-14B

AWQ), against same-model control forks. At the switch point the new model produces the next action *and* the continuation, so the two change together; this is our F contrast, not G or H. Its 359 control forks had 0 success flips; we claim no noise advantage. Restoring a state, replacing one action and letting the original policy continue is established: it locates decisive errors (Zhang et al., 2025), attributes failures through a do-intervention on one step (Shah, 2026), and turns alternative actions into training signal (Li et al., 2026a,b). This is the analogue of our G, and the factorial design itself is standard. None of the 32 closest works we checked crosses the action with the continuation controller at a precision-induced divergence; that crossing, which puts G, H and F on the same states and yields the $(\Delta_{\mathrm{Q}}, \Delta_{\mathrm{B}})$ table, bit-exact replay acceptance and the precision setting are what we add.

Precision routing and its proxies. DMR (Li et al., 2026c) learns a per-step BF16/W4 router from BF16-versus-quantized KL supervision and episode reward. Precision sensitivity has also been localised to phases (Lu et al., 2026) and task types (Zhang et al., 2026). We do not propose a competing router. We measure what a router would harvest, namely the intervention gains G, H and F at given states, and we test whether a Q-visible proxy predicts them. For quantized reasoning models, Lotfi et al. (2026) trace damage to high-entropy, high-KL token positions where markers such as "wait" are over-sampled; §7 tests the analogue for agent actions.

Evaluation noise and user simulators. Prefix caching changed multi-turn tool-use trajectories in 36.2% of episodes at 16-bit and 75.0% at 4-bit (Patodiya, 2026); agentic evaluations vary from run to run (Bjarnason et al., 2026; Chen et al., 2026), and with the user model (§6).

3 Crossed Same-State Intervention

3.1 Setting

Tasks and agents. We use τ^2-bench in text mode. The development panel has 58 retail and 27 airline tasks plus 80 telecom development tasks. A further 34 telecom tasks form a holdout that no agent ran before its protocol was frozen. The agents are Qwen3-4B-Instruct (q4) and Qwen3-8B with thinking disabled (q8). The user simulator is one greedy Qwen3-30B-A3B-Instruct model shared by all agents. Retail and airline success is the final-database match; telecom uses the official task-basis success. Without a user simulator, we also run BFCL multi-turn and ALFWorld, once per item (Appendix A).

Quantization and serving. Q is a simulated (fake-quantized) checkpoint served with BF16 compute on the same path as B. The main cells are GPTQ (Frantar et al., 2023) and AWQ (Lin et al., 2024) at W4 and a KV8 cache; stress cells are GPTQ-W3, RTN-W3 and a Q/K-rotated KV4 cache. B and Q run on vLLM with batch-invariant kernels and deterministic compilation, in separate server processes, so no prefix cache is shared across weights. A *block* is one closed-loop run of Q over the task pool under one prompt nonce, a random session-identifier line in the system prompt. Nonce blocks are the replicate unit.

3.2 Frozen states and estimands

Frozen state. Before Q's decision t in a recorded trajectory, we restore the state s_t. It comprises the full message history; the environment and user-device databases, rebuilt by strict replay of every recorded tool call; the user simulator's visible conversation; and the used step and error budget. B and Q each generate a full action from the same request, and the prompt token ids both servers receive must be identical. The state is written to disk before any branch runs.

Crossed intervention. $Y_{\mathrm{XY}} \in \{0, 1\}$ is task success when the action at s_t comes from X and controller Y takes over from the next step to the end, with X, Y $\in \{\mathrm{B}, \mathrm{Q}\}$. The action is executed by the real environment or answered by the real simulator and consumes one step; all four branches start from the same frozen state with the original remaining budget. Each escalation option is a paired contrast with QQ:

$$
\begin{aligned}
G &= Y_{\mathrm{BQ}} - Y_{\mathrm{QQ}} \quad \text{(this step only)},\\
H &= Y_{\mathrm{QB}} - Y_{\mathrm{QQ}} \quad \text{(from the next step)},\\
F &= Y_{\mathrm{BB}} - Y_{\mathrm{QQ}} \quad \text{(from here on)}.
\end{aligned}
$$

Their paired differences compare the options directly: $G - H = Y_{\mathrm{BQ}} - Y_{\mathrm{QB}}$, $F - G = Y_{\mathrm{BB}} - Y_{\mathrm{BQ}}$ and $F - H = Y_{\mathrm{BB}} - Y_{\mathrm{QB}}$. The effect of swapping the action can be read under either controller: $\Delta_{\mathrm{Q}} = Y_{\mathrm{BQ}} - Y_{\mathrm{QQ}}$ (Q continues; equal to G) and $\Delta_{\mathrm{B}} = Y_{\mathrm{BB}} - Y_{\mathrm{QB}}$ (B continues). Each takes values in $\{-1, 0, +1\}$, which gives a 3×3 table per

state population. On its diagonal, the action swap has the same effect under both controllers. Off the diagonal, the interaction $I = \Delta_Q - \Delta_B$ is nonzero. The two corners $(+1, -1)$ and $(-1, +1)$ are *strict reversals*: the better action under one controller is the worse one under the other. The contrasts measure effects on task success, not which action is correct.

Rescue and damage. For the current-action swap, a *rescue* is $(Y_{BQ}, Y_{QQ}) = (1, 0)$ and a *damage* is $(0, 1)$. With rates r and d, the outcome turnover is $M = r + d$ and the net gain is $r - d$.

3.3 Replay acceptance

Causal reading requires exact replay. Before any development outcome was analysed, an engineering gate on real divergence states in retail, airline and telecom passed every check; one B→B field, the official reward, was missing from older trajectories and was recomputed from the stored reward information before the verdict. Q→Q no-ops matched in 20/20 states and B→B no-ops in 10/10, in every prompt-token hash, message, database hash, termination, reward and budget counter. In production, every fork task re-ran the QQ no-op on a 10% sample of its states, selected by a hash of the state identifier and therefore independent of outcomes; a single mismatch would have quarantined the task's whole output. All 924 sampled states of the development analysis passed, as did 328/328 sampled Bank U states; the Bank U gate passed a further 135/135 no-ops. Serving closes the prefix-cache channel (Patodiya, 2026).

3.4 State banks

Bank P: first tool-structure divergences. For each task, block and quantized cell, we walk along Q's trajectory to the first decision at which a_B and a_Q differ in kind (tool call, text, stop), tool name, call count or order, or normalised arguments. **Differences in text only are skipped**: two different natural-language replies, or identical calls with different accompanying text, are not Bank P states, and the walk continues past them.

Bank U: ordinary states. Bank U draws one decision per task uniformly, with the draw fixed by a hash, in three nonce blocks (f01–f03); its manifest was frozen before any generation. It includes *all* full language actions: every state runs all four branches unless a_B and a_Q are token-identical, and no effect is imputed. U1 states come from each quantized cell's own trajectories. U2B (BF16 trajectories) and U2Q (Q trajectories) give every cell of a model the identical state (a common-controller check is in Appendix G).

3.5 Denominators and inference

We use three denominators and never mix them. (1) *Full pool*: every task × block unit of a cell, with units that have no Bank P state counted as zero. r, d, M and the net gain $G_{all} = r - d$ use it (§4). (2) *Divergent states*: the Bank P states of one cell, as a ratio of block sums. G, H, F and the 3×3 table use it (§5); G_{all} is G diluted by the share of units without divergence. (3) *Bank U*: one state per task, task-equal, per sub-bank (§6).

All tasks of a block share its nonce, and all branches share each state. Every within-cell contrast is therefore paired and *identified* for that cell's state population. Comparisons across cells (q4 vs. q8, AWQ vs. GPTQ, W4 vs. KV8) compare different state populations, because each cell selects its own first divergences, so they are *descriptive*. Only complete blocks enter; a block with any missing unit leaves the analysis. The primary interval treats the task pool as fixed: a block bootstrap with at least 10 complete blocks, otherwise a t interval over block means. Stress cells have 3 blocks, so their t intervals rest on two degrees of freedom and can be wider than the crossed intervals. A crossed task × block bootstrap is the interval for generalising to new tasks. Bank U intervals are t over its three block means.

4 R1: What Replacing the Current Action Rescues and Damages

Two-sided turnover, small net gain. On the τ^2-bench panel (Figure 2; all cells in Table 2; 85 tasks × 20 complete blocks), replacing a W4 agent's first tool-structure-divergent action changes the task outcome in $M = 15.5$–20.9% of units. Rescues and damages are of similar size, $r = 6.8$–11.4 and $d = 7.5$–10.1 pp, so the net gain G_{all} lies between -1.9 and $+2.9$ pp. Two fixed-pool intervals exclude zero (negative for q4 GPTQ-W4, positive for q8 GPTQ-W4); no new-task interval does. Pooled over the four W4 cells, G_{all} is $+0.7$ $[-0.3, +1.6]$ pp with $M = 18.6\%$. KV8 cells turn over fewer outcomes ($M = 6.5$–8.1%; a descriptive comparison across cells), and four of the five stress cells reach W4-like turnover ($M = 18.4$–20.4%) with only three blocks; q8 RTN-W3, whose own

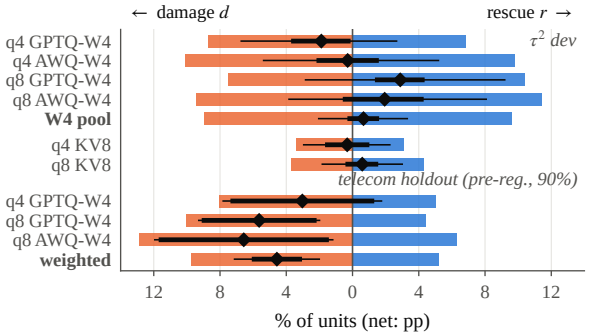


Figure 2: Rescue r and damage d (% of all task×block units) and net $G_{\text{all}} = r - d$ (pp; diamonds, negative to the left). Thick/thin: fixed-pool/new-task 95% (τ^2-bench dev.); block-t/crossed 90% (pre-registered holdout; "weighted" is its decisional endpoint). All cells: Table 2.

success is low, turns over 6.7%. G_{all} is a local replacement effect, not an end-to-end success comparison (Appendix B).

Turnover concentrates in tasks. In an exploratory, retrospective analysis of the development data, the top 20% of tasks hold 57–71% of all flips in the W4 and KV8 cells, and task-level turnover and its direction repeat across disjoint halves of the nonce blocks (Appendix H). On unseen telecom tasks, the pre-registered test that per-task turnover repeats across nonce halves (S2) passed (§6); S2 tests repeatability only, and task difficulty could explain it.

Same-precision baseline. To separate precision-specific turnover from generic trajectory sensitivity, we repeat the fork with Q′, a second GPTQ-W4 calibration draw of the same model, in place of B (q4 and q8 GPTQ-W4, the same Q episodes and tasks, a protocol frozen before any outcome). On the first K blocks, Q′ diverges from Q in x% of units and B in y%, and $\Delta M = M - M' = \Delta M$ [CI] pp (fixed pool; primary). By the frozen rule (an interval covering zero with $|\Delta M| \leq 3$ pp reads as generic sensitivity), reading.

5 R2: Where to Replace

On average over W4 cells, no option has a detectable gain. Pooled over the four W4 cells (Figure 3), $G = +0.5$, $H = +0.2$ and $F = +0.7$ pp, and every interval covers zero, as do their differences.

Within cells, the paired gains differ. In q8 GPTQ-W4, $G = +3.3$ [+1.6, +5.0] pp exceeds $H = +0.4$ ($G - H = +2.9$ [+1.3, +4.7]), and $F = +5.4$ exceeds G ($F - G = +2.1$ [+0.2, +4.0]). In q8 AWQ-W4, $G - H = +3.4$ [+1.5, +5.2] and $F - G = -1.5$ [−4.2, +0.8]. In q4 GPTQ-W4, $G = -2.9$ and $G - H = -3.6$ [−6.6, −0.6]. In the stress cell q8 RTN-W3 (three blocks), $H = +15.3$ and $F = +18.4$ against $G = -0.4$ ($G - H = -15.7$ [−27.5, −3.9]). In q4 KV8, B continuing after Q's action lowers success ($H = -2.5$ [−4.5, −0.5], unadjusted). After Holm adjustment over the four core W4 cells, $G - H$ stays below .05 in both q8 cells ($p = .013$, .012) but not in q4 GPTQ-W4 ($p = .069$). These within-cell orderings are exploratory: they hold on the fixed 85 tasks but are not confirmed on new tasks (Table 5) or under a second simulator (§6), and cells select different states.

A BF16 reference action can lower success. In q4 GPTQ-W4, B's action lowers success under Q's continuation on the fixed task pool ($G = -2.9$ [−5.7, −0.2] pp, unadjusted; the new-task interval covers zero), with a similar point estimate under B's ($\Delta_{\text{B}} = -3.2$ [−6.4, +0.2]).

The action's consequence often depends on who continues. In 26.9% [25.7, 28.1] of pooled W4 states, the action swap has a different effect under B's continuation than under Q's (Figure 4; pooled $(\Delta_{\text{Q}}, \Delta_{\text{B}})$ tables in Appendix Figure 5). Per W4 cell the share is 22.9–30.5%, and in the KV8 pool it is 15.3% [13.2, 17.3]. Because replay is exact, each such state shows a real interaction on the success scale, not replay noise; whether it is specific to precision is what the same-precision baseline tests (below and §4). Cohen's κ between Δ_{Q} and Δ_{B}, against a marginal-preserving re-pairing of states, is 0.29 in the W4 pool; it measures how well the two effects agree across states, not whether the interaction is significant (Appendix E.1). Strict sign reversals are rare: 3.2% of pooled W4 states, and 1.2–4.2% of states in each W4 and KV8 cell, or 8–14% of the off-diagonal mass. The rest is an effect under one controller and none under the other.

Controller dependence without an LLM user. In a post hoc analysis of BFCL multi-turn and ALFWorld, which have no simulated user (one run per item; Figure 4, Table 7), the off-diagonal share is 4.7–17.1% in the four BFCL W4 cells (pool 10.3%) and 32.8–40.0% in the two q8 ALFWorld W4 cells (pool 36.4%). Controller dependence

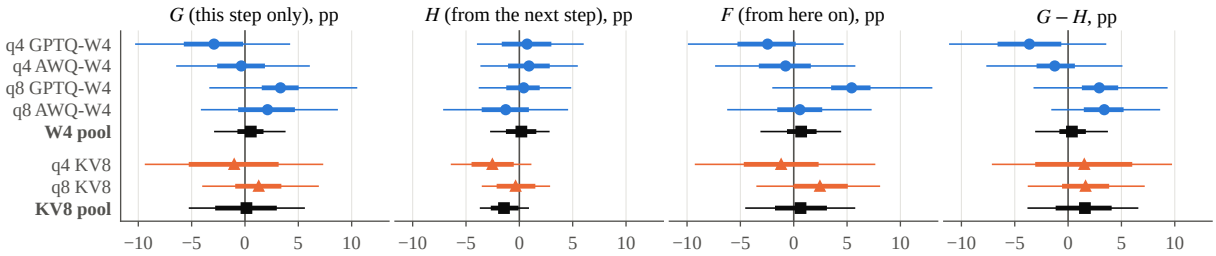


Figure 3: Replacement gains G, H, F and $G - H$ per core cell (pp; each cell's Bank P divergent states). Thick: fixed-pool 95% (block bootstrap, 20–22 blocks); thin: new-task 95% (crossed task×block). Squares: equal-weight pools. Within a row, contrasts are paired; between rows, descriptive. Stress cells, $F - G$, $F - H$ and (Δ_Q, Δ_B) summaries: Table 3.

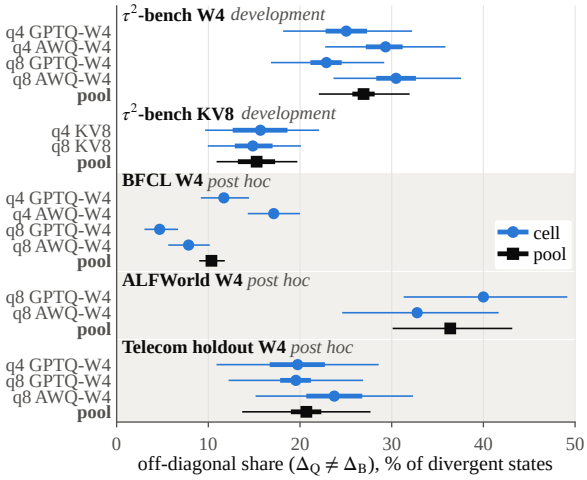


Figure 4: Same action, different value: per-cell share of divergent states with $\Delta_Q \neq \Delta_B$. Thick: fixed-pool 95% (development: block bootstrap; holdout: block t); thin: new-task 95% (crossed; BFCL/ALFWorld: item bootstrap). Pools: equal weight (holdout: pre-registered weights). Shaded groups are post hoc. Cross-cell comparisons are descriptive.

therefore does not require an LLM user simulator, and its size differs by environment.

Why the mean interaction is near zero. $E[I]$ is the difference between the swap's average effects under the two controllers, whereas the off-diagonal share is $P(I \neq 0)$. In the W4 pool, states where the swap helps more under Q ($I > 0$) and under B ($I < 0$) are about equally frequent (13.6% and 13.4%), so $E[I] = +0.0$ $[-1.9, +2.0]$ pp although 26.9% of states depend on the controller (new-task interval 22.1–32.0%). The value of a fixed replacement policy is $E[G]$, $E[H]$ or $E[F]$, not $E[I]$. All 16 outcome patterns are in Appendix D.

Same-precision baseline (descriptive). With Q′ in place of B (§4), the off-diagonal share is off′ against off with B on the same blocks. Each design selects its own states, so this comparison is descriptive. On units that diverge in both designs the shares are (a), and (b) where both diverge at the same decision.

6 R3: Do Sampling and Environment Change the Answer?

Ordinary states turn over less, mostly through text under our simulator. At ordinary states, replacing Q's action changes the outcome less often than at first tool-structure divergences (a descriptive comparison of different state populations, with no interval on the difference). Over blocks f01–f03 (task-equal; t intervals over block means), M per sub-bank is 7.8–9.0% for the q4 W4 cells in U1 and U2B, 4.7–9.0% for the q8 W4 cells and at most 4.7% for KV8. The net gain lies within ±2.8 pp (U1, U2B) and ±5.0 pp (U2Q), and every interval covers zero (Appendix G). Identical actions never change the outcome, as determinism implies. After deduplication, 150 of the 208 outcome-changing cell×state rows (≈72.1%) are class (ii), the same tool structure with different text; 26 involve a different tool name, call count or argument (Appendix Table 10). The rows are not independent states: the three blocks repeat the same tasks, and each U2B/U2Q state appears once per cell of a model. This is a *composition share*: per instance, text, tool/argument and act/talk differences flip the outcome in 14.4%, 18.2% and 18.9% of rows. Text dominates because it is common (1045 of 1357 differing rows), not because each text difference matters more. Among simple content features, only an above-median length difference between the two texts co-occurs with outcome change (in all four U1/U2B × model groups); question, confirmation-request and new-entity flags show no consistent associa-

tion. These are untested regular-expression features (Appendix G). Bank P cannot see these differences by design.

Sensitivity to the user simulator. Text actions act through the user simulator, and changing the simulator moves τ-bench outcomes by 9–15 points (Seshadri et al., 2026; Jain and Sandhu, 2026). We froze a control before running it: the 39 deduplicated class-(ii) flips of block f01 and 39 matched non-flip states. Their BQ and QQ branches were each re-run three times with a different user simulator, Sonnet 5.5, answering through a file relay from Claude Code subagents (reproducibility limits in the Limitations). Judged by the sign of the replicate mean, only 7 of the 39 flips keep their direction (95% CI 0.075–0.335). Six reverse, and 26 vanish (15 both succeed, 2 both fail, 9 ties). Among the matched non-flip states, 11 of 39 become flips (0.15–0.45). Replicates of the same arm disagree in 41% of flip arms and in 21–26% of control arms. **The Bank U text-difference outcome labels did not reproduce reliably under this alternative relay setting.** The control changes three things at once (simulator model, sampling, relay interface), so it cannot say which of them changed the labels (scope and sensitivity: Appendix G).

Sensitivity of the four-arm results. Before running, we sampled 80 Bank P states without outcome information (20 per W4 cell, stratified by domain and block) and re-ran all four arms five times each with Sonnet 5.5 as the user, one isolated single-request call per user turn (7896 calls; all integrity checks passed). Per-state arm outcomes track the original single runs (correlation 0.63–0.68). Pooled over cells, $G = -3.0$ [-10.2, $+4.8$], $H = +5.5$ [$+0.0$, $+11.2$] and $G - H = -8.5$ [-14.8, -2.8] pp under Sonnet, against $G - H = -2.5$ [-11.2, $+6.2$] on the same states under the original simulator (change -6.0 [-16.2, $+4.2$]). With 20 states per cell, this control cannot resolve the few-pp contrasts of Table 3 (on this sample even the original simulator gives $G - H = 0.0$ in both q8 W4 cells), so it neither confirms nor reverses R2's within-cell orderings. Per-cell estimates under both simulators are in Appendix Figure 6, and an exploratory mean squared interaction under Sonnet is in Appendix E.2.

Unseen tasks: telecom holdout. Before any holdout unit was run, we froze a confirmatory protocol (Appendix K): 34 unseen telecom tasks $\times$ 12 fresh nonce blocks $\times$ 3 cells (q4 GPTQ-W4, q8 GPTQ-W4, q8 AWQ-W4, weighted 1/2, 1/4, 1/4); primary endpoint G_{all} on a crossed 90% interval against ± 3 pp; secondaries two-sidedness and repeatability of task fragility. All 12 blocks were complete; 1200 of the 1224 units were analysed, after 2 structural task$\times$cell exclusions (Appendix K). The primary endpoint is $G_{\text{all}} = -4.57$ pp, crossed 90% interval [-7.17, -1.96] (block-t [-6.08, -3.05]). The pre-registered readings define only a positive gain, a gain above 3 pp and a small effect, so the decisional reading is *inconclusive*. Outside those readings, the interval lies entirely below zero: on these unseen tasks, replacing the first tool-structure-divergent action with BF16's lowered average success. Both secondaries pass. Rescues and damages are each above zero (Figure 2; $r = +5.21$ [$+3.30$, $+7.11$], $d = +9.77$ [$+6.45$, $+13.10$] pp), and task fragility repeats across independent nonce halves ($\rho = 0.59$, permutation $p = .0001$). Strata are in Appendix Table 17. On 80 telecom development tasks (6 blocks; fixed-pool 95%), the two q8 holdout cells give $G_{\text{all}} = +2.7$ [-4.7, $+10.1$] (GPTQ-W4) and -0.6 [-4.9, $+3.7$] pp (AWQ-W4), against -5.6 and -6.6 on the holdout; q4 has no telecom development forks. A post hoc R2 analysis of the 1180 divergent states (Table 18) gives $G = -4.6$ [-7.8, -1.5], $H = -2.8$ [-6.3, $+0.6$], $F = -2.5$ [-6.2, $+1.3$], $G - H = -1.8$ [-4.8, $+1.1$] and mean $I = -5.0$ [-9.6, -0.4] pp (crossed 95%), unlike the near-zero development W4 pool, with 20.7% of states off the diagonal (Figure 4). This G conditions on divergence and nearly equals G_{all} because 1180 of 1200 units diverge.

No user simulator: BFCL and ALFWorld. BFCL multi-turn and ALFWorld have no simulated user; each item runs once, and intervals are item-level. Replacing a W4 agent's first tool-structure-divergent action changes the outcome in 5.2–24.6% of items, in both directions, so two-sided turnover also occurs without a simulated user (ALFWorld: q8 only). The net gain covers zero except for BFCL q4 AWQ-W4, where it is -3.8 [-6.5, -1.3] pp (Table 7). A within-Qwen-family scale sensitivity check (Qwen3-14B, GPTQ-W4, BFCL, all four arms; a single configuration, not second-family evidence) gives $G = +5.7$ [$+2.9$, $+8.7$], $H = +0.4$ pp and 6.4% controller-dependent states (471 divergent). Con-

troller dependence on these benchmarks is reported with R2 (§5, Appendix E.3).

7 R4: Can Beneficial Replacements Be Found in Advance?

G, H and F are oracle quantities that require running B and all four branches; a deployable rule sees only Q. Following Lotfi et al. (2026), we scored every Bank U f01 state at the token level (1140 cell $\times$ state rows, 58 flips), teacher-forcing both models on the generated ids of a_{B} and a_{Q}. We record truncated top-20 entropy, the top-1 margin and a top-20-union KL, not full entropy or KL. All analyses are exploratory, and f01 is development data.

Uncertainty is associated with divergence. Q's entropy at the start of its action separates states where B and Q act differently from states where they act identically, with AUC 0.66–0.91 per sub-bank and model. Over all ordinary states, flips also sit at lower Q margins (U1: AUC 0.31 [0.22, 0.40] for q4, 0.24 [0.13, 0.36] for q8; stratified).

Among divergent states, it does not reliably identify consequences. Restricting to states where the actions differ removes the "any difference at all" signal. In U1 the signal is not detectable: no Q-visible statistic has an interval excluding 0.5 (21 and 9 flips; f01). Only small sub-banks keep a signal: q4 U2B (mean entropy over a_{Q}, AUC 0.73 [0.53, 0.89]) and q4 U2Q (entropy and margin at the action start, AUC 0.90 and 0.10; 10 states, 4 flips). At the first divergent token, pooled entropy and KL do not separate flips from non-flips except in q4 U2Q (KL AUC 0.80 [0.65, 0.93], entropy 0.25 [0.07, 0.48]; 4 flips). Per cell, two further unadjusted intervals exclude 0.5, in opposite directions (U1 q4 GPTQ-W4 entropy 0.30; U2B q4 KV8 KL 0.10). No class of first divergent token reproduces the marker enrichment of Lotfi et al. (2026). The largest q4 enrichment is ordinary word choice (rate ratio 1.89, Fisher $p = .019$, not corrected). On development data, uncertainty is associated with *where* quantization changes the action more strongly than with *which* changes matter.

A frozen trigger, tested on held-out blocks. We froze a Q-visible trigger on f01 before any f02/f03 state was scored: escalate one decision when the minimum Q margin over a_{Q} is low, with per-cell thresholds at 5, 10 and 20% escalation rates (Appendix I). On held-out f02/f03 (1020 U1 rows, 69 turnover states) it escalated 50, 114 and 222 states. Net gains were $+4$, $+2$ and $+5$ (rescues/damages 6/2, 7/5, 16/11) against matched-count random expectations of $+0.21$, $+0.98$ and $+2.38$ (exact one-sided hypergeometric $p = .0345$, .417, .262). It captured 8, 12 and 27 turnover states against 3.48, 7.52 and 14.30 expected ($p = .0182$, .0613, .0002). With no level fixed in advance, net gain passes a nominal one-sided .05 only at the 5% rate, and not after Holm correction. The frozen rule thus finds outcome-changing states but has no significant net advantage over random selection, and its held-out blocks reuse the development tasks.

Headroom. A hindsight action-replacement oracle with the same per-cell counts nets $+3.33$, $+3.73$ and $+3.73$ pp. This headroom is positive by construction: the oracle picks only rescues (padding with neutral states) and never a damage. The rule captures 4/34, 2/38 and 5/38 of that net. A menu oracle and a post-hoc $\mathrm{KL}(\mathrm{B}\|\mathrm{Q})$ row (not Q-only; no significant net gain, Holm $p = 1.0$) are in Appendix I (Figure 7).

8 Discussion

When to escalate. Average success, action divergence and replacement gain disagree here. The four W4 cells have end-to-end differences we can neither distinguish from zero nor bound within ± 3 pp (Appendix B), yet replacing their first tool-structure-divergent action turns over many units (§4). Escalation can also hurt (q4 KV8 H, §5; telecom holdout, §6). No replacement unit was best in every cell (§5), so G, H and F should be measured per configuration and environment. [Same-precision reading: one sentence.] We recommend reporting r, d and M next to any net gain, each with its denominator, and keeping fixed-pool and new-task intervals apart.

Bank P and Bank U measure different quantities. Bank P skips text-only differences by design, yet these make up 150 of the 208 outcome-changing rows at ordinary states (§6), and their labels did not reproduce under a second simulator. Forking only at tool-structure divergences misses most text-driven turnover; including text actions requires a simulator control before their consequences are reported as a property of the action.

Limitations

What we do not claim. We do not claim success equivalence between BF16 and W4 agents; equivalence tests at ± 3 pp on new-task intervals fail. G, H and F are oracle quantities, not the value of a deployable policy (§7). We do not claim a W$\times$KV interaction, and we do not present static precision restoration as a method (Appendix L).

One model family, two sizes. All crossed forks use Qwen3 models: 4B and 8B throughout, plus a single Qwen3-14B configuration on BFCL. Effects may differ at other scales and in other families. End-to-end runs also exist for Qwen3-14B and Qwen3-32B; apart from that one Qwen3-14B configuration they have no crossed forks, and we draw no conclusion from them (Appendix F). A time-boxed second-family check (Llama-3.1-8B-Instruct, GPTQ-W4, BFCL multi-turn) was stopped at its gate: BF16 solved 5 of 120 gate items, so the four arms carried almost no outcome variance, and it yields no second-family evidence.

Simulated quantization. Quantized checkpoints are dequantized and served with BF16 compute on a deterministic path. Deployed low-bit kernels change the numerics and add serving variation that we deliberately remove. We make no claims about latency, memory or deployed kernels, and the cost of a BF16 fallback is outside our scope.

One user simulator; two small controls. All τ^2-bench episodes use a single greedy simulator, and continuation effects include its replies. Both second-simulator controls replace only the user after a fixed history, cover 78 and 80 states, and say nothing about real users. The Bank P control has 20 states per cell, too few to confirm or refute few-pp contrasts. Neither control can be reproduced draw for draw. The replacement user is a hosted Sonnet 5.5 model sampled with its default settings, without temperature or seed control. In the Bank U control, Claude Code subagents answered many relay requests in one context, so a responder could see other arms or replicates of the same conversation. The Bank P control removed this exposure by issuing one isolated, single-request call per user turn. The same-precision baseline (§4) is our only reference for how much turnover and controller dependence a non-precision perturbation produces; it covers GPTQ-W4 on the development panel only.

LLM annotation. Process labels in Appendix J come from LLM annotators (Sonnet 5.5, two independent instances plus an adjudicator) and rule-based checks. No human annotation has been done. Agreement among instances of one model family shows reproducibility, not correctness.

Small confirmation and partial banks. The pre-registered holdout covers 34 telecom tasks and three cells. Its pre-registered readings had no negative category, so a clearly negative estimate could only be read as inconclusive; its negative estimate is established only for telecom, these three cells and this simulator. Most development results come from retail and airline, so the holdout confounds a domain shift (telecom) with generalisation to new tasks. On 80 telecom development tasks, the two q8 holdout cells show no detectable net effect (§6), so the telecom domain alone does not produce a detectable drop; but their 6-block intervals are wide and q4 has no telecom development forks, so the holdout drop may reflect heterogeneity among its 34 tasks as much as a domain effect. Bank P intervenes only at the first tool-structure divergence per task and block. Bank U samples one state per task and block and has three blocks, so its block intervals rest on two degrees of freedom. Prompt nonces define replicate blocks; they are a researcher-chosen perturbation, not a deployment distribution.

Ethics Statement

This work involves no crowdworkers and no personal data. All users are simulated by an LLM on public benchmark tasks, and all agents are openly released Qwen3 checkpoints. The process annotations were produced by LLMs on synthetic benchmark conversations. The main risk we see is over-reading. A small forked-state gain on one simulator and a fixed task pool does not certify a quantized agent for deployment, and our end-to-end comparisons do not establish equivalence. Deployment decisions in consequential settings should not rest on these measurements alone.

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A Setup Details

Splits. The development split has 58 retail, 27 airline and 80 telecom tasks. The holdout has 34 telecom tasks, and its pre-registration hash is recorded in the split file. The telecom split was fixed from τ^2-bench's 114-task base split by a stratified, salted hash before any run. The external benchmarks use BFCL multi-turn (base, miss_func, miss_param and long_context, 197 items each) and ALFWorld (134 out-of-distribution games). Items were selected by hash in manifests committed before any model run.

Success endpoints. For retail and airline we use the final-database match with normal termination. This is not τ^2-bench's official reward, which additionally scores natural-language or communicate checks. Telecom uses the official deterministic task-basis success. Context overflow and unparseable output count as failures. Infrastructure outages are missing, never failures, and a block with a missing unit leaves the primary analysis.

Fork mechanics. The forced action is returned by a fork agent for the one step that follows h_t. A tool call is executed by the real environment, and text is answered by the real simulator, which never reuses observations from the original trajectory. τ^2-bench resets step and error counters

when it starts from a history. The fork restores them to the values used at t, so the remaining budget equals the original. Every max_steps and too_many_errors episode on the checked blocks is reproduced by the recount. The served user simulator is Qwen3-30B-A3B-Instruct in every episode; a user-model label stored in the fork state files is outdated and should be ignored.

B End-to-End Success Comparison

The replacement gains G_{all}, G, H and F are not end-to-end comparisons. Table 1 gives the end-to-end comparison, computed separately on the same checkpoints, the same tasks and the same 20 nonce blocks as the forks.

Cell	Q / BF16 (%)	Q−BF16 [95% CI]	NI 3 pp	TOST ±3 pp
q4 GPTQ-W4	40.0 / 35.8	+4.2 [−2.4, +10.5]	pass	fail
q4 AWQ-W4	36.8 / 35.8	+0.9 [−5.8, +7.5]	fail	fail
q8 GPTQ-W4	30.1 / 30.4	−0.2 [−7.7, +6.8]	fail	fail
q8 AWQ-W4	31.5 / 30.4	+1.2 [−5.3, +7.6]	fail	fail
pool, all four		+1.5 [−2.7, +5.6]	pass	fail

Table 1: End-to-end closed-loop success, quantized minus BF16 (pp). Units are paired on nonce, and the crossed task×block bootstrap is drawn jointly across cells. Non-inferiority (NI) at a 3 pp margin passes for q4 GPTQ-W4 and the all-four pool. Equivalence (TOST at ±3 pp) passes nowhere. These comparisons therefore support neither a difference nor equivalence.

C Full Tables for R1 and R2

Table 2 gives every cell of Figure 2, including the stress cells, with new-task intervals. Table 3 gives every cell of Figure 3, with $F - G$, $F - H$ and the $(\Delta_{\text{Q}}, \Delta_{\text{B}})$ summaries.

D All 16 Outcome Patterns and New-Task Intervals

Table 4 lists every four-branch pattern per cell; `0000` (39–58%) and 1111 (6–20%) dominate. Table 5 gives the new-task intervals for Table 3.

E Supplementary R2 Analyses

E.1 Marginal-preserving pairing agreement control

For each cell, observed agreement is the diagonal share of the $(\Delta_{\text{Q}}, \Delta_{\text{B}})$ table, $p_o = \sum_k P(\Delta_{\text{Q}} = k, \Delta_{\text{B}} = k)$, so the off-diagonal share is $1 - p_o = P(I \neq 0)$. The control re-pairs states independently within a cell while keeping both marginals, $p_e = \sum_k p_Q(k)\, p_B(k)$, and Cohen's $\kappa = (p_o - p_e)/(1 - p_e)$. Pools use the paper's equal weight per cell over common complete blocks, with the same bootstrap draws as Table 3. This is a pairing agreement control, not a no-interaction null and not a chance baseline for controller dependence (§5). In the W4 pool the control expects 38.0% [36.7, 39.0] off-diagonal against the observed 26.9%, so $\kappa = 0.29$ [0.27, 0.32] (new-task [0.22, 0.36]); in the KV8 pool it expects 33.6% ($\kappa = 0.55$). The effects under the two controllers agree more than random cross-state pairing would. On BFCL and ALFWorld the observed shares also lie below the control (expected 16.2–32.4% and 42.9–46.4% in the W4 cells; κ 0.47–0.71 and 0.14–0.24; Table 7), and on the telecom holdout the pooled 20.7% compares with 25.7% expected ($\kappa = 0.19$; Table 18).

Cell	units	div.	r	d	M	G_{all} [fixed pool]	[new tasks]
q4 GPTQ-W4	85×20	1102	6.8	8.7	15.5	−1.9 [−3.7, −0.1]	[−6.8, +2.7]
q4 AWQ-W4	85×20	1439	9.8	10.1	19.9	−0.3 [−2.2, +1.6]	[−5.4, +5.2]
q8 GPTQ-W4	85×20	1473	10.4	7.5	17.9	+2.9 [+1.4, +4.4]	[−2.9, +9.2]
q8 AWQ-W4	85×20	1562	11.4	9.5	20.9	+1.9 [−0.6, +4.3]	[−3.9, +8.1]
W4 pool	85×20		9.6	9.0	18.6	+0.7 [−0.3, +1.6]	
q4 KV8	85×22	593	3.1	3.4	6.5	−0.3 [−1.7, +1.0]	[−3.0, +2.3]
q8 KV8	85×22	855	4.3	3.7	8.1	+0.6 [−0.4, +1.6]	[−1.9, +3.0]
q4 GPTQ-W3^s	85×3	232	9.0	10.2	19.2	−1.2 [−7.0, +4.7]	[−8.6, +5.9]
q4 KV4-rot^s	85×3	233	11.8	7.1	18.8	+4.7 [−4.1, +13.5]	[−3.1, +12.9]
q8 GPTQ-W3^s	85×3	242	10.2	8.2	18.4	+2.0 [−12.5, +16.4]	[−6.3, +11.8]
q8 KV4-rot^s	85×3	239	11.8	8.6	20.4	+3.1 [−14.5, +20.8]	[−5.9, +12.9]
q8 RTN-W3^s	85×3	255	3.1	3.5	6.7	−0.4 [−3.8, +3.0]	[−4.7, +3.9]

Table 2: Replacing Q's first tool-structure-divergent action while Q continues, τ^2-bench retail/airline panel (pp). Denominator: the full pool of tasks × complete blocks; units without a Bank P state count as zero. div. = Bank P states. $G_{\text{all}} = r - d$ is a local replacement effect, *not* the end-to-end success difference (Appendix B). Fixed pool: block bootstrap (≥10 blocks) or t over block means; new tasks: crossed task×block bootstrap. ^{s}stress cell, 3 blocks.

E.2 Mean squared interaction under the second simulator (exploratory)

On the 80 states of the Bank P simulator control (§6), each arm has $n_a = 5$ replicates under Sonnet 5.5. Per state, $\hat{I}_s = \bar{Y}_{\text{BQ}} + \bar{Y}_{\text{QB}} - \bar{Y}_{\text{BB}} - \bar{Y}_{\text{QQ}}$ and $U_s = \hat{I}_s^2 - \sum_a \bar{Y}_a(1 - \bar{Y}_a)/(n_a - 1)$, which is unbiased for I_s^2 if, given the state, the replicates of each arm are i.i.d. Bernoulli and the four arms are independent (no shared cache or seed). Pooled with equal weight over the four W4 cells, mean $\hat{I}_s^2$ is 0.110, the noise correction 0.060 and mean U_s 0.050, with 95% intervals [0.004, 0.107] over states and [0.001, 0.111] over task clusters. One cell is negative (q4 AWQ-W4, −0.025) and is kept, as are negative per-state values. The es-

Cell	states	G (this step)	H (next step on)	F (from here on)	$G - H$	$F - G$	$F - H$ ($= \Delta_B$)	off-diag. %	rev. %	mean I
q4 GPTQ-W4	1102	−2.9 [−5.7, −0.2]	+0.7 [−1.6, +3.0]	−2.5 [−5.3, +0.2]	−3.6 [−6.6, −0.6]	+0.5 [−1.2, +2.2]	−3.2 [−6.4, +0.2]	25.0	2.6	+0.3
q4 AWQ-W4	1439	−0.3 [−2.6, +1.9]	+0.9 [−1.1, +2.9]	−0.8 [−3.3, +1.6]	−1.3 [−3.0, +0.6]	−0.4 [−2.4, +1.5]	−1.7 [−3.6, +0.4]	29.3	3.1	+1.3
q8 GPTQ-W4	1473	+3.3 [+1.6, +5.0]	+0.4 [−1.2, +1.9]	+5.4 [+3.5, +7.2]	+2.9 [+1.3, +4.7]	+2.1 [+0.2, +4.0]	+5.0 [+3.2, +6.8]	22.9	2.7	−1.7
q8 AWQ-W4	1562	+2.1 [−0.6, +4.7]	−1.3 [−3.5, +0.9]	+0.6 [−1.5, +2.7]	+3.4 [+1.5, +5.2]	−1.5 [−4.2, +0.8]	+1.9 [−0.1, +3.8]	30.5	4.2	+0.3
W4 pool	5576	+0.5 [−0.7, +1.7]	+0.2 [−1.3, +1.6]	+0.7 [−0.6, +2.1]	+0.4 [−0.8, +1.7]	+0.2 [−1.0, +1.2]	+0.5 [−1.0, +2.1]	26.9	3.2	+0.0
q4 KV8	593	−1.0 [−5.3, +3.2]	−2.5 [−4.5, −0.5]	−1.2 [−4.7, +2.3]	+1.5 [−3.1, +6.0]	−0.2 [−3.5, +3.7]	+1.3 [−2.5, +4.9]	15.7	1.2	−2.4
q8 KV8	855	+1.3 [−0.9, +3.4]	−0.4 [−2.1, +1.5]	+2.5 [+0.0, +5.1]	+1.6 [−0.6, +3.9]	+1.2 [−0.5, +3.2]	+2.8 [+0.3, +5.5]	14.9	1.2	−1.5
KV8 pool	1448	+0.1 [−2.8, +3.0]	−1.4 [−2.7, −0.1]	+0.6 [−1.8, +3.1]	+1.6 [−1.2, +4.1]	+0.5 [−1.6, +2.8]	+2.1 [−0.2, +4.2]	15.3	1.2	−1.9
q4 GPTQ-W3^s	232	−1.3 [−7.7, +5.0]	+4.3 [−10.7, +19.4]	+1.3 [−1.9, +4.4]	−5.6 [−15.0, +3.7]	+2.6 [−0.6, +5.8]	−3.0 [−15.1, +9.1]	31.5	3.9	+1.7
q4 KV4-rot^s	233	+5.2 [−4.5, +14.9]	+9.4 [+5.1, +13.7]	+12.4 [+11.0, +13.9]	−4.3 [−18.3, +9.8]	+7.3 [−2.0, +16.5]	+3.0 [−1.9, +8.0]	33.0	1.7	+2.1
q8 GPTQ-W3^s	242	+2.1 [−13.3, +17.2]	+0.4 [−7.4, +8.2]	+2.1 [−7.4, +11.4]	+1.7 [−9.1, +12.3]	+0.0 [−11.0, +11.1]	+1.7 [−0.1, +3.4]	23.1	4.5	+0.4
q8 KV4-rot^s	239	+3.3 [−15.4, +21.9]	−0.8 [−18.2, +16.3]	+4.6 [−16.0, +25.0]	+4.2 [+2.5, +5.8]	+1.3 [−1.8, +4.3]	+5.4 [+0.8, +10.0]	31.0	4.6	−2.1
q8 RTN-W3^s	255	−0.4 [−3.8, +3.0]	+15.3 [+0.7, +29.9]	+18.4 [+16.7, +20.1]	−15.7 [−27.5, −3.9]	+18.8 [+13.8, +23.9]	+3.1 [−13.0, +19.2]	29.0	0.4	−3.5

Table 3: Three replacement gains and their paired differences on each cell's Bank P states (pp; same denominator in every column). $G = Y_{BQ} - Y_{QQ}$, $H = Y_{QB} - Y_{QQ}$, $F = Y_{BB} - Y_{QQ}$. Off-diag.: share of states with $\Delta_Q \neq \Delta_B$; rev.: strict reversals. Fixed-pool 95% intervals (block bootstrap over 20–22 complete blocks; t over 3 blocks for ^{s}stress cells, which cannot be pooled). New-task intervals are in Table 5. Within a row, contrasts are paired and identified; between rows, descriptive.

$(Y_{BB}, Y_{BQ}, Y_{QB}, Y_{QQ})$	q4 AWQ	q4 GPTQ-W3	q4 GPTQ	q4 KV4r	q4 KV8	q8 AWQ	q8 GPTQ-W3	q8 GPTQ	q8 KV4r	q8 KV8	q8 RTN-W3
0000	634 (44.1%)	105 (45.3%)	472 (42.8%)	92 (39.5%)	286 (48.2%)	767 (49.1%)	130 (53.7%)	766 (52.0%)	113 (47.3%)	497 (58.1%)	141 (55.3%)
0001	55 (3.8%)	7 (3.0%)	29 (2.6%)	3 (1.3%)	14 (2.4%)	66 (4.2%)	2 (0.8%)	36 (2.4%)	8 (3.3%)	15 (1.8%)	3 (1.2%)
0010	57 (4.0%)	13 (5.6%)	54 (4.9%)	13 (5.6%)	11 (1.9%)	38 (2.4%)	8 (3.3%)	34 (2.3%)	10 (4.2%)	15 (1.8%)	20 (7.8%)
0011	58 (4.0%)	6 (2.6%)	71 (6.4%)	4 (1.7%)	37 (6.2%)	30 (1.9%)	9 (3.7%)	49 (3.3%)	3 (1.3%)	39 (4.6%)	0 (0.0%)
0100	50 (3.5%)	6 (2.6%)	36 (3.3%)	9 (3.9%)	8 (1.3%)	53 (3.4%)	8 (3.3%)	45 (3.1%)	13 (5.4%)	20 (2.3%)	1 (0.4%)
0101	21 (1.5%)	4 (1.7%)	17 (1.5%)	4 (1.7%)	11 (1.9%)	36 (2.3%)	7 (2.9%)	12 (0.8%)	5 (2.1%)	5 (0.6%)	2 (0.8%)
0110	22 (1.5%)	5 (2.2%)	14 (1.3%)	2 (0.9%)	2 (0.3%)	33 (2.1%)	4 (1.7%)	16 (1.1%)	2 (0.8%)	5 (0.6%)	1 (0.4%)
0111	41 (2.8%)	5 (2.2%)	21 (1.9%)	7 (3.0%)	13 (2.2%)	51 (3.3%)	6 (2.5%)	31 (2.1%)	7 (2.9%)	7 (0.8%)	6 (2.4%)
1000	51 (3.5%)	8 (3.4%)	32 (2.9%)	19 (8.2%)	16 (2.7%)	67 (4.3%)	10 (4.1%)	75 (5.1%)	16 (6.7%)	28 (3.3%)	32 (12.5%)
1001	23 (1.6%)	4 (1.7%)	15 (1.4%)	2 (0.9%)	5 (0.8%)	33 (2.1%)	7 (2.9%)	24 (1.6%)	9 (3.8%)	5 (0.6%)	0 (0.0%)
1010	18 (1.3%)	5 (2.2%)	13 (1.2%)	9 (3.9%)	4 (0.7%)	17 (1.1%)	5 (2.1%)	17 (1.2%)	3 (1.3%)	3 (0.4%)	20 (7.8%)
1011	36 (2.5%)	9 (3.9%)	33 (3.0%)	9 (3.9%)	8 (1.3%)	32 (2.0%)	3 (1.2%)	19 (1.3%)	2 (0.8%)	11 (1.3%)	6 (2.4%)
1100	44 (3.1%)	3 (1.3%)	51 (4.6%)	9 (3.9%)	41 (6.9%)	43 (2.8%)	10 (4.1%)	82 (5.6%)	9 (3.8%)	46 (5.4%)	2 (0.8%)
1101	36 (2.5%)	7 (3.0%)	27 (2.5%)	3 (1.3%)	9 (1.5%)	38 (2.4%)	4 (1.7%)	23 (1.6%)	1 (0.4%)	11 (1.3%)	1 (0.4%)
1110	51 (3.5%)	9 (3.9%)	15 (1.4%)	10 (4.3%)	7 (1.2%)	65 (4.2%)	4 (1.7%)	34 (2.3%)	6 (2.5%)	10 (1.2%)	4 (1.6%)
1111	242 (16.8%)	36 (15.5%)	202 (18.3%)	38 (16.3%)	121 (20.4%)	193 (12.4%)	25 (10.3%)	210 (14.3%)	32 (13.4%)	138 (16.1%)	16 (6.3%)
n	1439	232	1102	233	593	1562	242	1473	239	855	255

Table 4: All 16 four-branch outcome patterns per cell: count (share of the cell's Bank P states). Pattern digits are $(Y_{BB}, Y_{BQ}, Y_{QB}, Y_{QQ})$. KV4r: Q/K-rotated KV4; W3 and KV4r cells are stress cells with 3 blocks.

timate is the mean squared probability-scale interaction over these 80 frozen states. It is not the share of states with interaction and not a variance, it is not comparable with the deterministic 26.9% of §5, and we make no significance claim.

E.3 R1 and R2 estimands on BFCL and ALFWorld (supplementary)

Table 7 gives the R1 quantities of §6 (r, d, M and G_{all} over all items, non-divergent items counting as zero) and applies R2 to the external benchmarks, from the raw four-arm outcomes on divergent items (one run per item; item bootstrap within benchmark category). The R2 columns are post hoc and not part of any pre-registered analysis. In the KV4 cells $Y_{QQ} = 0$ on every state, so Δ_Q is almost constant, their off-diagonal mass is H and κ is near 0 by construction. In BFCL q4 GPTQ-W3, the base category was dropped whole by the QQ no-op rule, leaving 591 items and 540 states. Off-diagonal shares are 4.7–17.1% in BFCL W4 cells (pool 10.3%) and 32.8–40.0% in q8 ALFWorld W4 (pool 36.4%); pairing κ is in Appendix E.1. The ALFWorld GPTQ-W3 stress cell has $H = +38.5$ and $G - H = -34.6$ [−43.5, −25.4] pp; KV4 cells reduce to H.

F Coverage Matrix

Table 8 lists the completed crossed-intervention coverage behind every result. It counts tasks, states and complete blocks after the strict admission rule.

G Bank U per Sub-Bank

Table 9 gives the per-state turnover behind §6, over blocks f01–f03. Exact action agreement is 0.43–0.83 per cell and sub-bank.

Text content. Over f01–f03, an above-median relative length difference between the two texts co-occurs with outcome change in U1 and U2B of both models (flipping pairs among above- vs. below-median pairs: q4 U1 30/154 vs. 15/155; q8 U1 22/101 vs. 6/101; q4 U2B 33/138 vs. 11/139; q8 U2B 16/99 vs. 9/100). In f01 alone this was visible only for q4. Question, confirmation-request and new-entity flags show no consistent associa-

Cell	G	H	F	$G-H$	$F-G$	$F-H$
q4 GPTQ-W4	$[-10.3, +4.2]$	$[-4.0, +6.0]$	$[-9.9, +4.7]$	$[-11.2, +3.6]$	$[-4.7, +6.1]$	$[-11.1, +4.4]$
q4 AWQ-W4	$[-6.5, +6.1]$	$[-3.7, +5.5]$	$[-7.4, +5.8]$	$[-7.7, +5.1]$	$[-6.2, +5.4]$	$[-8.7, +4.7]$
q8 GPTQ-W4	$[-3.3, +10.5]$	$[-3.8, +4.9]$	$[-2.0, +13.0]$	$[-3.2, +9.3]$	$[-3.0, +7.1]$	$[-2.0, +12.5]$
q8 AWQ-W4	$[-4.1, +8.7]$	$[-7.1, +4.6]$	$[-6.2, +7.3]$	$[-1.6, +8.6]$	$[-7.6, +4.2]$	$[-3.8, +7.5]$
W4 pool	$[-2.9, +3.8]$	$[-2.7, +2.8]$	$[-3.1, +4.4]$	$[-3.1, +3.7]$	$[-2.8, +3.1]$	$[-3.5, +4.5]$
q4 KV8	$[-9.4, +7.3]$	$[-6.4, +1.1]$	$[-9.3, +7.7]$	$[-7.1, +9.7]$	$[-5.7, +5.3]$	$[-6.7, +9.5]$
q8 KV8	$[-4.0, +6.9]$	$[-3.5, +2.9]$	$[-3.5, +8.1]$	$[-3.8, +7.2]$	$[-2.5, +5.0]$	$[-3.1, +8.7]$
KV8 pool	$[-5.3, +5.6]$	$[-3.7, +0.9]$	$[-4.5, +5.8]$	$[-3.8, +6.6]$	$[-2.8, +3.9]$	$[-3.0, +7.3]$
q4 GPTQ-W3^s	$[-9.7, +6.6]$	$[-6.6, +14.5]$	$[-6.8, +9.4]$	$[-15.0, +4.0]$	$[-5.2, +10.5]$	$[-12.2, +6.6]$
q4 KV4-rot^s	$[-3.4, +14.2]$	$[+2.1, +17.0]$	$[+3.0, +21.6]$	$[-13.5, +7.0]$	$[-1.3, +17.1]$	$[-6.2, +12.2]$
q8 GPTQ-W3^s	$[-6.7, +12.1]$	$[-7.2, +8.1]$	$[-6.6, +11.8]$	$[-7.7, +11.2]$	$[-9.1, +9.1]$	$[-6.8, +11.0]$
q8 KV4-rot^s	$[-6.6, +13.8]$	$[-11.2, +8.5]$	$[-6.6, +15.9]$	$[-3.3, +12.2]$	$[-7.2, +9.9]$	$[-2.5, +14.6]$
q8 RTN-W3^s	$[-4.7, +3.9]$	$[+5.9, +25.9]$	$[+9.4, +28.2]$	$[-24.7, -7.1]$	$[+10.2, +27.8]$	$[-7.1, +13.3]$

Table 5: New-task generalisation: crossed task×block bootstrap 95% intervals for the estimates of Table 3 (pp; same Bank P states). These answer whether the effect would hold for new tasks drawn like the panel; the fixed-pool intervals of Table 3 treat the 85 tasks as fixed. ^{s}stress cell, 3 blocks.

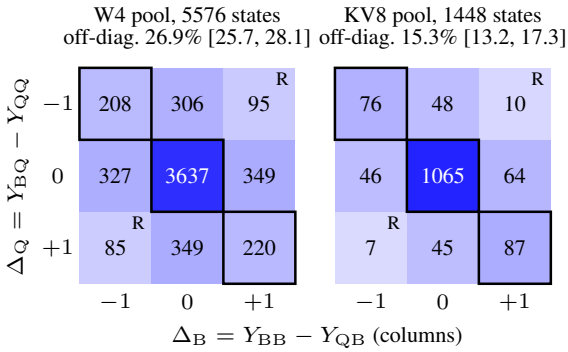


Figure 5: Does the action swap have the same effect under both controllers? Counts of (Δ_Q, Δ_B) over the Bank P states of the W4 cells (left) and the KV8 cells (right). Boxed: diagonal. R: strict reversals. Shading: share of states (power scale). Headers give the off-diagonal share as the equal-weight average over cells, the aggregation used throughout the paper except for the state-weighted menu oracle of Appendix I, with fixed-pool 95% intervals.

Cell	states	off-diag. %	exp. %	κ [fixed pool]	κ [new tasks]
q4 GPTQ-W4	1102	25.0 [22.8, 27.3]	40.5	0.381 [0.325, 0.437]	[0.245, 0.520]
q4 AWQ-W4	1439	29.3 [27.2, 31.2]	38.5	0.238 [0.215, 0.261]	[0.138, 0.343]
q8 GPTQ-W4	1473	22.9 [21.1, 24.5]	36.3	0.369 [0.324, 0.409]	[0.235, 0.492]
q8 AWQ-W4	1562	30.5 [28.3, 32.6]	36.8	0.171 [0.128, 0.213]	[0.075, 0.274]
W4 pool	5576	26.9 [25.7, 28.1]	38.0	0.291 [0.265, 0.318]	[0.223, 0.359]
q4 KV8	593	15.7 [12.7, 18.6]	36.2	0.567 [0.487, 0.643]	[0.412, 0.705]
q8 KV8	855	14.9 [12.9, 17.0]	31.1	0.522 [0.462, 0.584]	[0.386, 0.656]
KV8 pool	1448	15.3 [13.2, 17.3]	33.6	0.546 [0.487, 0.604]	[0.436, 0.652]
q4 GPTQ-W3^s	232	31.5 [10.5, 52.6]	36.1	0.129 [−0.258, 0.534]	[−0.049, 0.347]
q4 KV4-rot^s	233	33.0 [17.6, 48.3]	38.0	0.131 [−0.016, 0.280]	[−0.034, 0.298]
q8 GPTQ-W3^s	242	23.1 [6.0, 40.2]	36.4	0.364 [0.190, 0.548]	[0.199, 0.532]
q8 KV4-rot^s	239	31.0 [19.0, 43.0]	37.7	0.179 [0.098, 0.251]	[0.032, 0.327]
q8 RTN-W3^s	255	29.0 [24.6, 33.5]	28.6	−0.016 [−0.233, 0.200]	[−0.101, 0.142]

Table 6: Pairing agreement control per cell (Bank P divergent states). Off-diag.: observed share with $\Delta_Q \neq \Delta_B$ (fixed-pool 95%); exp.: share expected under marginal-preserving within-cell re-pairing. Pools: equal weight per cell. ^{s}Stress cells (3 blocks; t intervals; flagged, never pooled). In q8 RTN-W3, $\Delta_Q = 0$ on 93.3% of states, so p_e is close to p_o and $\kappa \approx 0$.

tion. The features are regular expressions, and this is descriptive.

Bank U simulator control: sensitivity and scope. Dropping two mechanically flagged replies leaves the reading of §6 unchanged (6, 6 and 25 of 37; Table 11). In either direction, 13 of the 39 original flip states flip again, against 11 of 39 new flips at matched non-flip states; this does not show that the two groups are equivalent. The relay's responders may have seen other requests. The control's scope is narrow: fixed histories from the original simulator, a stochastic relay rather than API Sonnet or real users, block f01 only and three replicates.

Shared fragility across quantizers (a lead). On the 255 shared q4 U2B states of f01–f03, cell pairs co-flip 21 times, against about 3.3 expected under independence, and 16 of the 21 co-flips go in the same direction. They come from 15 distinct states, 3 of which flip under all three cells. On q8 U2B, where every f01 flip was single-cell, 10 states now flip under two or more cells (14 pairwise co-flips against about 2.8 expected; 10 of 14 in the same direction). Joint flips almost never share an identical quantized action across cells (0 of 21 in q4, 2 of 14 in q8). We still read this as a lead toward *state-level* fragility, where different actions at the same state flip the outcome, and not yet as a finding: the co-flips are counted on one fixed set of states per block, with no correction.

A common controller (Q0). On the frozen 25% subsample of U2 in f01–f03, also continued by a common controller Q0, B's action leads to the identical outcome in every cell on all 146 shared states, which checks determinism. Turnover under Q0 is not consistently above or below the native turnover on the same rows (q4 AWQ-W4 U2B

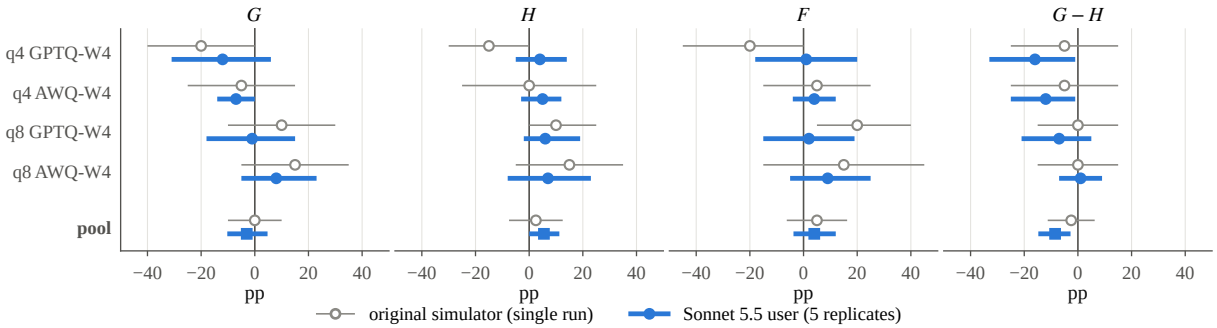


Figure 6: Bank P simulator control: G, H, F and $G - H$ on the 80 sampled Bank P states (20 per W4 cell), under the original greedy simulator (single runs; open circles) and with Sonnet 5.5 as the user (five replicates per arm; filled). pp; 95% intervals over states; pool: equal weight per cell. The sample is too small to test few-pp contrasts (§6).

	R1: all items					R2: divergent items								
Cell	items	r	d	M	G_{all}	states	G	H	F	$G - H$	off-diag. %	exp. %	κ	rev. (n)
BFCL q4 GPTQ-W4	788	6.3	4.3	10.7	+2.0 [−0.3, +4.3]	573	+2.8 [−0.4, +5.9]	+1.4 [−0.7, +3.5]	+4.2 [+0.9, +7.2]	+1.4 [−1.8, +4.5]	11.7	27.4	0.574	3
BFCL q4 AWQ-W4	788	5.7	9.5	15.2	−3.8 [−6.5, −1.3]	642	−4.7 [−7.8, −1.5]	−1.1 [−3.6, +1.4]	−1.4 [−4.8, +1.7]	−3.6 [−6.7, −0.2]	17.1	32.4	0.471	8
BFCL q8 GPTQ-W4	788	2.9	2.3	5.2	+0.6 [−0.9, +2.3]	533	+0.9 [−1.3, +3.3]	+1.5 [+0.0, +3.2]	+2.1 [−0.4, +4.6]	−0.6 [−3.1, +2.0]	4.7	16.2	0.710	1
BFCL q8 AWQ-W4	788	4.1	2.7	6.7	+1.4 [−0.3, +3.3]	548	+2.0 [−0.4, +4.7]	+1.8 [+0.0, +3.7]	+3.6 [+0.9, +6.5]	+0.2 [−2.6, +3.1]	7.8	19.9	0.607	0
BFCL W4 pool					+0.1 [−1.0, +1.2]		+0.3 [−1.2, +1.8]	+0.9 [−0.2, +2.0]	+2.1 [+0.3, +3.9]	−0.6 [−2.3, +1.0]	10.3			
ALFWorld q8 GPTQ-W4	134	9.0	12.7	21.6	−3.7 [−11.2, +4.5]	115	−4.3 [−13.3, +5.0]	+7.0 [−2.6, +16.7]	+9.6 [−0.9, +19.7]	−11.3 [−20.9, −1.8]	40.0	46.4	0.137	4
ALFWorld q8 AWQ-W4	134	10.4	14.2	24.6	−3.7 [−11.9, +4.5]	119	−4.2 [−13.6, +4.9]	+5.0 [−2.6, +12.2]	+4.2 [−5.9, +13.7]	−9.2 [−18.8, +0.0]	32.8	42.9	0.237	3
ALFWorld W4 pool					−3.7 [−9.7, +1.9]		−4.3 [−11.1, +2.2]	+6.0 [−0.5, +12.6]	+6.9 [−2.1, +15.5]	−10.3 [−17.7, −2.9]	36.4			
BFCL q4 GPTQ-W3^s	591	6.1	3.6	9.6	+2.5 [+0.2, +5.1]	540	+2.8 [+0.2, +5.5]	+5.2 [+2.9, +7.7]	+8.9 [+5.7, +12.2]	−2.4 [−5.7, +0.7]	13.0	23.3	0.443	1
BFCL q8 GPTQ-W3^s	788	5.2	4.1	9.3	+1.1 [−1.0, +3.3]	696	+1.3 [−1.1, +3.7]	+1.4 [−0.1, +2.9]	+3.4 [+0.9, +6.0]	−0.1 [−2.6, +2.4]	8.3	20.4	0.592	1
BFCL q4 KV4^s	788	0.3	0.0	0.3	+0.3 [+0.0, +0.6]	787	+0.3 [+0.0, +0.6]	+2.9 [+1.9, +4.1]	+20.3 [+17.6, +23.0]	−2.7 [−3.9, −1.5]	20.5	20.9	0.020	0
BFCL q8 KV4^s	788	0.3	0.0	0.3	+0.3 [+0.0, +0.6]	788	+0.3 [+0.0, +0.6]	+1.8 [+1.0, +2.7]	+14.5 [+12.1, +17.0]	−1.5 [−2.5, −0.6]	14.7	15.2	0.029	0
ALFWorld q8 GPTQ-W3^s	134	6.7	3.0	9.7	+3.7 [−1.5, +9.7]	130	+3.8 [−1.5, +9.8]	+38.5 [+29.8, +46.9]	+42.3 [+33.6, +50.8]	−34.6 [−43.5, −25.4]	30.8	34.1	0.098	0
ALFWorld q8 KV4^s	134	0.0	0.0	0.0	+0.0 [+0.0, +0.0]	134	+0.0 [+0.0, +0.0]	+38.1 [+30.6, +45.5]	+53.0 [+44.8, +61.2]	−38.1 [−45.5, −30.6]	26.9	26.9	0.000	0

Table 7: R1 and R2 estimands on BFCL multi-turn and ALFWorld (no LLM user simulator; one run per item; 95% item-bootstrap intervals). R1: rescue r, damage d, turnover M (% of all items) and net G_{all} (pp), non-divergent items counting as zero. R2 (supplementary, exploratory, post hoc; pp, divergent items): Off-diag.: share with $\Delta_{\text{Q}} \neq \Delta_{\text{B}}$; exp. and κ: marginal-preserving pairing agreement control; rev.: strict reversals. Pools: equal weight per cell. ^{s}Stress and unrotated KV4 cells; in KV4 $Y_{\text{QQ}} = 0$ on every state, so the table reduces to H.

8/58 vs. 5/58; q8 AWQ-W4 U2B 4/52 vs. 5/52; q4 KV8 U2B 1/58 vs. 2/58), so the f01 observation that Q0 turnover is at least the native one does not hold over three blocks. With 17–58 states per cell and sub-bank, this is too small to say whether the controller changes which actions matter.

H Concentration of Turnover in Tasks

This appendix completes the exploratory concentration result of §4 (development data, 85 tasks × 20 blocks). The per-task dispersion of M is 4.0–8.7 times its binomial null in the W4 and KV8 cells, and the top 20% of tasks hold 57–71% of all flips. We split the 20 blocks into two disjoint halves of 10 nonces. Task-level M correlates across halves (Spearman 0.52–0.86). Tasks that were net-rescue in one half have a mean net gain in the other half that is 14–51 pp higher than that of net-damage tasks, in all 12 cell-directions. A Freedman–Lane test conditions on measured baselines: QQ success propensity and BF16 success, plus trajectory length in a post-hoc variant. With them, one half's task rate still predicts the other's in all 6 configurations × 2 directions for both M and G (Holm-adjusted). This is a *retrospective conditional association* on viewed development data. It is carried by retail. Airline alone gives 8 of 12 for M (5 after Holm), and 3 of 12 under a heteroscedasticity-robust wild bootstrap. Direction persistence beyond flip opportunity holds more often for rescues than for damages, in 11 versus 6 of the 12 cell-directions. Different quantizers flip the same tasks: the task-identity null is rejected in 7 of 8 quantizer pairs. When both flip, however, they agree in direction only 42–81% of the time. This overlaps with the stable failure set of Jang et al. (2026); unlike that work, we record the direction of each flip.

I Trigger Headroom and a KL Reference Row

Table 12 compares the frozen trigger of §7 (frozen rule, unchanged) with two references on the same 1020 f02/f03 U1 rows, which hold 38 rescues and 31 damages. The *matched action-replacement or-*

Cell	τ^2-bench retail (Bank P)	τ^2-bench airline (Bank P)	telecom dev (Bank P)	Bank U (retail / airline)	BFCL / ALFWorld
q4 GPTQ-W4	58 t · 727 st · 20/20 b	27 t · 375 st · 20/20 b	holdout only	3 b · 393 / 177 rows	788 items · 573 st / –
q4 AWQ-W4	58 t · 986 st · 20/20 b	27 t · 453 st · 20/20 b	–	3 b · 393 / 177 rows	788 items · 642 st / –
q4 KV8	58 t · 412 st · 22/22 b	27 t · 181 st · 22/22 b	–	3 b · 393 / 177 rows	– / –
q8 GPTQ-W4	58 t · 1017 st · 20/20 b	27 t · 456 st · 20/20 b	80 t · 477 st · 6/6 b; holdout	3 b · 393 / 177 rows	788 · 533 st / 134 · 115 st
q8 AWQ-W4	58 t · 1087 st · 20/20 b	27 t · 475 st · 20/20 b	80 t · 480 st · 6/6 b; holdout	3 b · 393 / 177 rows	788 · 548 st / 134 · 119 st
q8 KV8	58 t · 585 st · 22/22 b	27 t · 270 st · 22/22 b	–	3 b · 393 / 177 rows	– / –
q4 GPTQ-W3	58 t · 162 st · 3/3 b	27 t · 70 st · 3/3 b	–	–	591 · 540 st† / –
q4 KV4-rot	58 t · 164 st · 3/3 b	27 t · 69 st · 3/3 b	–	–	– / –
q8 GPTQ-W3	58 t · 166 st · 3/3 b	27 t · 76 st · 3/3 b	–	–	788 · 696 st / 134 · 130 st
q8 KV4-rot	58 t · 165 st · 3/3 b	27 t · 74 st · 3/3 b	–	–	– / –
q8 RTN-W3	58 t · 174 st · 3/3 b	27 t · 81 st · 3/3 b	–	–	– / –
q4 / q8 KV4 (unrotated)	–	–	–	–	788 · 787 / 788 · 788 st; q8 ALFWorld 134 · 134 st
q14 GPTQ-W4	–	–	–	–	788 · 471 st / –

Table 8: Completed crossed-intervention coverage (t = tasks, st = states, b = complete/found blocks; BFCL and ALFWorld: items · states, one run). Bank U: blocks completed and state rows per domain; blocks f01–f03 are analysed (Appendix G). "holdout": the cell is in the telecom holdout (Appendix K). Qwen3-32B has end-to-end runs only; Llama-3.1-8B stopped at its fork gate (Limitations); the q14 (Qwen3-14B) BFCL row is a within-family scale check whose counts are pending. q4 RTN-W3 has no complete Bank P block and is omitted. The telecom development results (§6) use the 6 blocks listed. †The BFCL base category of q4 GPTQ-W3 was dropped by the QQ no-op rule (Appendix E.3). The end-to-end q4 AWQ-W3 run is incomplete (two airline episodes hit 600 s serving timeouts), and so are the q4 RTN-W2 stress runs (context-window overflows of the degenerate model and a truncated result file); neither is reported.

Cell	U1 (n=255)	U2B (n=255)	U2Q (n=60)
q4 GPTQ-W4	.090 (11/12)	.082 (10/11)	.083 (4/1)*
q4 AWQ-W4	.082 (7/14)	.078 (7/13)	.083 (3/2)
q4 KV8	.047 (4/8)	.039 (5/5)	.017 (0/1)
q8 GPTQ-W4	.047 (8/4)	.067 (9/8)	.050 (3/0)*
q8 AWQ-W4	.086 (14/8)	.090 (15/8)	.067 (3/1)
q8 KV8	.035 (5/4)	.031 (6/2)	0 (0/0)

Table 9: Bank U, blocks f01–f03: per-state turnover M (rescues/damages), task-equal (one state per task and block; 85/85/20 states per block). G lies within ± 0.028 in U1 and U2B and within ± 0.050 in U2Q, and every interval includes 0. Intervals are t intervals over the three block means and are wide (e.g. q4 GPTQ-W4 U1: .090 [.029, .151]). *GPTQ-W4's U2Q states are its own U1 states, with concordant outcomes on 20/20 duplicates per block; they are counted once in cross-sub-bank totals.

Model:sub	(i) identical	(ii) same tool structure, text	(iii) different tool or args	(iv) act vs. talk
q4:U1	399: 0	309: 45 (.146)	19: 1 (.053)	38: 10 (.263)
q8:U1	483: 0	202: 28 (.139)	47: 11 (.234)	33: 4 (.121)
q4:U2B	430: 0	277: 44 (.159)	24: 4 (.167)	34: 3 (.088)
q8:U2B	470: 0	199: 25 (.126)	40: 9 (.225)	56: 14 (.250)
q4:U2Q	108: 0	63: 10 (.159)*	5: 0	4: 1 (.250)
q8:U2Q	129: 0	33: 6 (.182)*	11: 1 (.091)	7: 0
unique rows	1943: 0	1045: 150 (.144)	143: 26 (.182)	169: 32 (.189)

Table 10: Which kind of action difference changes the outcome at ordinary states (Bank U, blocks f01–f03). Entries: "state rows: outcome-changing rows (flip rate)", pooled over the three cells of a model within a sub-bank. The last row counts each (cell, state) once: q4/q8 GPTQ-W4's U2Q states repeat their U1 states (* includes 5 and 3 repeated class-(ii) flips), so 216 flipping rows become 208. In block f01 alone the corresponding counts were 39 of 56 flips in class (ii), with rates .126/.109/.200 for (ii)/(iii)/(iv); the 58 f01 flips scored in §7 include the two duplicated U2Q rows.

acle selects, per cell, exactly as many rows as the frozen rule did, taking only rows with $G > 0$ and filling with $G = 0$ rows when a cell has too few rescues; it never selects a damage. It is hindsight selection, not a policy. At the 10% and 20% budgets every rescue fits. The *three-option menu oracle* $E[\max(G, H, F, 0)]$ is the share of states where some arm beats QQ. On Bank P divergent states it is 0.202 over the four W4 cells (state-weighted, the one exception to the paper's equal-weight pools) against 0.117 for the action alone (1127 vs. 654 of 5576 states), and 0.191 vs. 0.113 over the six core cells. It is hindsight selection that also allows different replacement types, with different costs, so it is not a same-cost deployment comparison and not comparable with the action rule.

The *KL reference row* is a post-hoc auxiliary baseline. Its score is a truncated $\mathrm{KL}(\mathrm{B}\|\mathrm{Q})$ over the union of both models' top-20 next-token sets plus a tail bucket, per a_Q token, teacher-forced on identical prompt ids. It needs a BF16 prefill of the prompt and a_Q at every decision (here 6709.3 prompt and 82.3 action tokens on average), so it is not Q-only. It captures turnover above chance (33 vs. 16.7 at 20%) but no significant net gain (Holm $p = 1.0$). Three aggregations over the a_Q tokens were fit on f01 only, with per-cell thresholds and the frozen rule's selection criterion; the maximum was chosen, and all of f02/f03 is reported. The two candidates not chosen on f01 look better

Analysis	flips	retained	rev.	vanished (s/f/tie)	ctrl	new flips (resc./dam.)
All replies	39	7 [0.07, 0.34]	6	26 (15/2/9)	39	11 [0.15, 0.45] (7/4)
Flagged dropped	37	6 [0.06, 0.32]	6	25 (15/2/8)	39	11 [0.15, 0.45] (7/4)

Table 11: Bank U simulator control (block f01; Sonnet 5.5 relay user, $R = 3$). Fates use the sign of the replicate-mean flip indicator; s/f: both arms succeed/fail. "Flagged dropped": the two arm replicates with a mechanically flagged reply are set to missing. Replicates of one arm disagree in 16/39 (BQ) and 16/39 (QQ) flip states and in 10/39 and 8/39 control states. Per single replicate, retained flips are 5, 7 and 4 of 39.

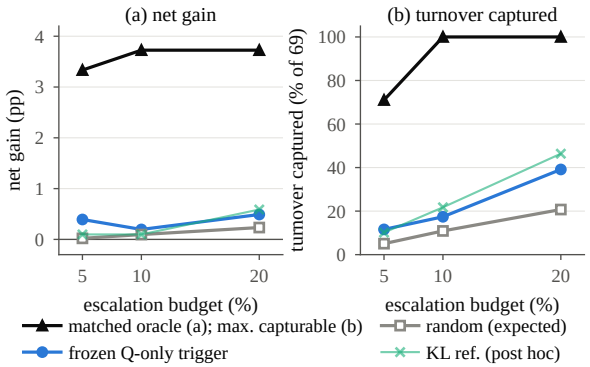


Figure 7: Frozen Q-only trigger by escalation budget on f02/f03 U1 (1020 rows; 38 rescues, 31 damages), against the matched action-replacement oracle (hindsight; same per-cell selection counts), the matched-count random expectation, and a faint post-hoc KL(B‖Q) reference (maximum per-token KL, ranked within the same per-cell counts; needs BF16). (a) Net gain, (rescues − damages)/1020, in pp. (b) Share of the 69 turnover states selected; here the black series is the most turnover any selection with the same per-cell counts can capture, $\sum_c \min(m_c, R_c + D_c)$ (derived from the per-cell counts), not the action oracle. Values: Table 12 (its KL row uses the KL rule's own counts).

on f02/f03 (e.g. the mean at 20%: net +9, Holm $p = .0852$), but selecting them now would be selection on validation data. Because the f02/f03 outcomes had been seen, the row is post hoc, and it is not a DMR reproduction, which trains a router.

J Controlled Injection and Process Audit

A natural explanation of R1 is a local-defect story: the quantized agent makes a local error, and the task fails unless the continuation recovers. Two analyses constrain this explanation without identifying a mechanism.

Controlled defect injection. We froze 30 ordinary states from q4 GPTQ-W4 trajectories, all from one development block. At each state we forced one of three actions as the next step: an INValid action, a matched VALid action, or a process-perturbation control (PP) that repeats the last successful read. Either B or Q then continued. The defect categories were wrong_id, wrong_tool and premature_write, with 10 states each. Every state passed a replay audit of 22 recorded fields (VAL×Q against the original episode). Three fields were not recorded: the counters right after the restore, the rendered-prompt hash sequence and the tool-schema hash sequence.

Result: two-sided turnover, no clear error-specific difference detected. Injected defects change the outcome in 7/30 states under B (4 rescues, 3 damages) and 4/30 under Q (2 and 2). The process-perturbation control changes 5/30 under both controllers (Table 13). The error-specific effect, INV minus PP, is 0.00 [−0.13, +0.13] under B and −0.03 [−0.20, +0.10] under Q. The controller dependence D is nonzero in 8 of 30 states, but its mean is +0.03 [−0.17, +0.27]. So injected defects cause heterogeneous two-sided turnover, and at $n = 30$ states we detect no clear difference from a generic process perturbation and no mean controller difference. The intervals are wide, so this does not show that errors and generic perturbations are equivalent.

Manipulation checks limit the categories. Two of 30 states failed a manipulation check: an injected write was rejected, and a valid reference itself returned "not found". Restricting to the 28 passing states leaves the reading unchanged. The wrong_tool category mostly produced argument errors: 8 of 10 injected calls failed with an unexpected-keyword error. We therefore call it a *tool-call/schema error perturbation*, not a wrong-tool test. premature_write is mostly failure→failure: the valid arm under Q already fails in 7/10 of these states, and turnover is 0/10 under both controllers. The event chain error → correction → related requirement satisfied completes in only 6 (B) and 7 (Q) of 29 injected errors, although 17 are corrected under each controller. Injections add tool errors (+0.43 under B, +0.57 under Q per branch).

Replication on q8 GPTQ-W4. The same design on 30 frozen q8 states (one development block; 10 per category) gives the same reading (Table 14). All 30 states pass the replay audit. Injected defects change the outcome in 5/30 states

Budget	same per-cell selection counts sel.	oracle r/d/net (pp)	frozen r/d/net (pp)	captured	KL reference (max. per-token KL; needs BF16; post hoc) sel.	net vs. random (p; Holm)	r/d	turnover vs. random (p)
5%	50	34/0/+34 (+3.33)	6/2/+4 (+0.39)	4/34	72	−1 vs. +0.45 (.8337; 1.0000)	5/6	11 vs. 4.58 (.0038)
10%	114	38/0/+38 (+3.73)	7/5/+2 (+0.20)	2/38	126	+1 vs. +0.82 (.5483; 1.0000)	8/7	15 vs. 7.54 (.0048)
20%	222	38/0/+38 (+3.73)	16/11/+5 (+0.49)	5/38	257	+3 vs. +1.59 (.3984; 1.0000)	18/15	33 vs. 16.73 (.0000)

Table 12: Frozen Q-only trigger against a matched action-replacement oracle (hindsight; same per-cell selection counts) and a post-hoc KL reference row, on f02/f03 U1 (1020 rows). Captured: frozen net over oracle net. Random: matched-count per-cell random-trigger expectation; p: exact one-sided hypergeometric, Holm over the three budgets.

Ctrl	Contrast	resc./dam.	turnover	net [95% CI]
B	INV vs VAL	4/3	7/30	+0.03 [−0.13, +0.20]
B	PP vs VAL	3/2	5/30	+0.03 [−0.10, +0.20]
B	INV vs PP	2/2	4/30	0.00 [−0.13, +0.13]
Q	INV vs VAL	2/2	4/30	0.00 [−0.13, +0.13]
Q	PP vs VAL	3/2	5/30	+0.03 [−0.10, +0.17]
Q	INV vs PP	2/3	5/30	−0.03 [−0.20, +0.10]
Controller dependence D ($\neq 0$ in 8/30 states)				+0.03 [−0.17, +0.27]

Table 13: Defect injection on 30 frozen q4 states (development, one block): per-state rescue/damage of one arm against another, under the BF16 (B) or quantized (Q) continuation controller. $D = [Y(\text{INV}, B) - Y(\text{VAL}, B)] - [Y(\text{INV}, Q) - Y(\text{VAL}, Q)]$. Task-clustered bootstrap intervals; all exact McNemar $p \geq 0.25$.

Ctrl	Contrast	resc./dam.	turnover	net [95% CI]
B	INV vs VAL	3/2	5/30	+0.03 [−0.10, +0.17]
B	PP vs VAL	4/2	6/30	+0.07 [−0.07, +0.23]
B	INV vs PP	1/2	3/30	−0.03 [−0.13, +0.07]
Q	INV vs VAL	3/2	5/30	+0.03 [−0.10, +0.17]
Q	PP vs VAL	5/0	5/30	+0.17 [+0.03, +0.30]
Q	INV vs PP	2/6	8/30	−0.13 [−0.30, +0.03]
Controller dependence D ($\neq 0$ in 6/30 states)				0.00 [−0.17, +0.17]

Table 14: Defect injection on 30 frozen q8 GPTQ-W4 states (development, one block), as in Table 13. Task-clustered bootstrap intervals; exact McNemar $p \geq 0.0625$.

under each controller (3 rescues, 2 damages), the process-perturbation control in 6/30 (B) and 5/30 (Q). The error-specific effect is −0.03 [−0.13, +0.07] under B and −0.13 [−0.30, +0.03] under Q; the latter interval still allows a sizeable error-specific harm, and D is nonzero in 6 of 30 states with mean 0.00 [−0.17, +0.17]. Under Q, the process-perturbation control itself rescues 5 states and damages none (+0.17 [+0.03, +0.30], McNemar $p = 0.0625$). Manipulation checks fail more often than in q4: 7 of the 10 premature writes failed the check (6 were rejected by the environment, 1 executed without changing the database), and 2 valid references returned "not found". On the 21 passing states, INV vs VAL turnover is 5 (B) and 4 (Q), and $D \neq 0$ in 5. The q8 premature_write category therefore mostly did not inject a write.

Process audit (LLM-annotated). We sampled 80 development fork states, 20 per outcome sign. Two independent Sonnet 5.5 instances annotated them blind, a third instance adjudicated, and rule-based programmatic labels were added. These annotations are a *process description* conditional on outcome-defined groups. They are not causal evidence, not prevalences and not human labels. Three patterns stand out (Table 15). First, local defects and final failure do not correspond one-to-one. In damage states, where Q's action led to success and B's to failure, Q's first-action defect is corrected later in 10 of 20 QQ branches. Second, whether a write executes does not separate the failing from the succeeding continuation of the same state (rescue 17 vs. 17; damage 15 vs. 14). Third, the failing continuation more often carries the rubric label `unintended_or_unconsented_change` (rescue: QQ 9 vs. BQ 2, paired difference +0.35 [+0.08, +0.57]; damage: 1 vs. 9, −0.40 [−0.62, −0.11]). In damage states, 7 of these 20 BQ branches carry the label on writes *with* recorded consent. That is the semantic "wrong change" part of the label, and it overlaps with the database-defined outcome by construction. Loops and budget exhaustion also mark failing branches (rescue: QQ 5 vs. BQ 0). We make no claim about consent. The programmatic consent field agrees poorly with the annotators, so we use it only as a mechanical row. Re-labelling 30 states with fresh instances of the same model family reproduces the key pattern.

J.1 Process audit table

The core "local defect" combines groundedness, argument validity and permission/consent judgements. The tool-validity field (inter-annotator $\kappa = 0.29$) is excluded from it. Mechanical fields agree

Label (source)	Comparison	rescue	damage	both succeed	both fail
`unintended_or_unconsented_change` (LLM)	QQ vs BQ	9 vs 2, $+0.35$ [$+0.08$, $+0.57$]	1 vs 9, -0.40 [-0.62, -0.11]	1 vs 1	5 vs 7, -0.10 [-0.35, $+0.17$]
write executed (programmatic)	QQ vs BQ	17 vs 17, 0.00 [-0.25, $+0.25$]	15 vs 14, $+0.05$ [-0.26, $+0.35$]	10 vs 9	11 vs 13
executed write w/o recorded consent (prog.)	QQ vs BQ	9 vs 5, $+0.20$ [-0.06, $+0.43$]	3 vs 3, 0.00 [-0.21, $+0.21$]	3 vs 2	4 vs 3
loop or budget exhaustion (LLM)	QQ vs BQ	5 vs 0, $+0.25$ [$+0.04$, $+0.47$]	0 vs 2, -0.10 [-0.30, $+0.08$]	1 vs 2	7 vs 4
action judged indefensible (LLM)	Q vs B action	7 vs 8, -0.05 [-0.37, $+0.28$]	12 vs 5, $+0.35$ [-0.05, $+0.65$]	10 vs 6	12 vs 9
local defect, core (LLM)	Q vs B action	4 vs 6, -0.10 [-0.33, $+0.14$]	12 vs 7, $+0.25$ [-0.09, $+0.53$]	6 vs 3	11 vs 7
QQ first-action defect corrected later (LLM)	single branch	4/20 [0.08, 0.42]	10/20 [0.30, 0.70]	4/20	7/20

Table 15: Process audit of 80 development fork states, 20 per outcome sign (rescue: $Y_{\mathrm{BQ}} = 1, Y_{\mathrm{QQ}} = 0$; damage: $Y_{\mathrm{BQ}} = 0, Y_{\mathrm{QQ}} = 1$). **LLM-annotated** by Sonnet 5.5 (two independent blind instances and an adjudicating instance), plus rule-based programmatic labels; no human labels. Counts are first vs. second of $n = 20$ states. Intervals are within-state paired differences with Newcombe 95% CIs; single-branch rows use Wilson 95% intervals. This is a process description conditional on outcome-defined groups: not causal, not prevalence. The label name is the rubric category; we make no claim about consent.

with the programmatic verifier (write executed vs. the rubric's irreversible-change field, $\kappa = 0.99$). The consent fields agree only moderately ($\kappa = 0.44$ on writes both see).

K Holdout Pre-Registration

The protocol was frozen before any holdout unit was run, and its hash was recorded in the split manifest. It fixes the following design:

- tasks, cells, weights and nonce blocks;
- two fixed halves of six blocks each;
- an outcome-blind repair rule, with reserve blocks used only for infrastructure repair;
- the decisional crossed interval (two-way ANOVA moments, normal 90%);
- the readings: positive if $L > 0$, above 3 pp if $L > 3$, small if the interval lies inside ± 3 pp, otherwise inconclusive;
- the secondary families S1 (both $r > 0$ and $d > 0$) and S2 (persona-stratified permutation, 10,000 draws, no outcome screen);
- six stratum families, with Holm correction within each.

The analysis script refuses holdout data unless the hash matches. Two task-cell pairs were excluded as structural, and the complete-case analysis is identical. Table 16 gives the confirmatory result. Table 17 gives all six stratum families (Holm within family), which are descriptive. Every level estimate is ≤ 0, and the Holm-significant levels are negative: step-budget T2 and T3, mobile-data issues (-9.09 pp) and the fewest-faults tercile (-6.37 pp).

Post-hoc R2 estimands on the holdout. Table 18 is a supplementary, descriptive output computed after the confirmatory analysis, with the same admission and units. It is not part of the pre-registered confirmation and changes neither the primary endpoint nor its reading. Its G is a ratio over divergent states, whereas G_{all} in Table 16 averages over all admitted units with non-divergent units counted as 0.

Cell (weight)	units	excl.	G	r	d	ρ_c
q4 GPTQ-W4 (1/2)	396	1	-3.03 [-7.86, $+1.80$]	$+5.05$	$+8.08$	0.668
q8 GPTQ-W4 (1/4)	408	0	-5.64 [-9.33, -1.94]	$+4.41$	$+10.05$	0.374
q8 AWQ-W4 (1/4)	396	1	-6.57 [-11.98, -1.15]	$+6.31$	$+12.88$	0.644
Weighted (G_{all})			-4.57 [-7.17, -1.96]	$+5.21$	$+9.77$	0.588

Table 16: Telecom holdout, pre-registered analysis (34 tasks $\times$ 12 blocks; pp; crossed 90% intervals). Per-cell rows are descriptive; the weighted row is decisional (G_{all}, reading: inconclusive) and S1/S2 (both pass). Excl.: structurally excluded tasks (decision/trajectory count mismatch at search).

L Withdrawn, Negative and Supplementary Results

W$\times$KV interaction (not supported). A first q4 analysis found a W4$\times$KV4 interaction of -14.9 [-23.8, -6.0] pp ($k = 3$). Its KV16 reference lacked the Q/K rotation used for KV4, so rotation and quantization were confounded. With a matched rotated reference, the q4 interaction is -9.0 [-30.6, $+12.5$]. The pre-registered q8 replication gives $+2.1$ [-5.8, $+10.1$] over 6 blocks, crossed [-8.2, $+12.4$]. The replication is not supported, and we make no W$\times$KV claim.

Static precision restoration (appendix only). We tested whether restoring selected modules to higher precision recovers a damaged RTN-W3 model. We report this line as a mechanism stress test, not a method. In RTN-W3 the gains largely tracked the number of restored bits. A random MLP arm also recovered much of the gap, and no arm beat whole-model GPTQ-W4 at matched stor-

Family	Level	n	G [crossed 90%]	p	Holm p
step budget[a]	T1	138/129/110	-0.32 [-1.57, $+0.94$]	.6771	.6771
	T2	124/126/162	-2.30 [-3.88, -0.72]	.0165	.0495
	T3	123/144/124	-1.94 [-3.35, -0.54]	.0229	.0495
persona	Hard	11	-5.99 [-12.23, $+0.26$]	.1147	.3355
	None	12	-3.65 [-7.42, $+0.13$]	.1118	.3355
	Easy	11	-4.36 [-9.19, $+0.48$]	.1385	.3355
BF16 difficulty	0	20	-2.35 [-4.09, -0.62]	.0257	.0772
	low	11	-4.54 [-8.90, -0.19]	.0860	.1720
	high	3	-18.75 [-38.00, $+0.50$]	.1091	.1720
task length	T1	9	-5.09 [-9.81, -0.38]	.0755	.1511
	T2	8	-0.78 [-4.46, $+2.90$]	.7270	.7270
	T3	17	-6.17 [-10.49, -1.84]	.0189	.0567
issue type	mms	14	-1.15 [-3.33, $+1.04$]	.3880	.4434
	mobile data	11	-9.09 [-14.98, -3.20]	.0111	.0333
	service	9	-4.40 [-10.32, $+1.52$]	.2217	.4434
faults	T1	19	-6.37 [-9.91, -2.83]	.0030	.0091
	T2	10	-1.94 [-7.58, $+3.69$]	.5704	.5704
	T3	5	-2.92 [-5.18, -0.66]	.0337	.0674

Table 17: Telecom holdout strata: the six pre-registered families (descriptive; Holm within each family over its three levels). pp. For task-level families, n is the number of tasks and G is G_{all} over the tasks in the level. [a]Step budget (remaining steps at the divergence) is unit-level: n is the number of divergent units per cell (q4 GPTQ-W4/q8 GPTQ-W4/q8 AWQ-W4), and G is the level's contribution to G_{all}, so the three levels sum to G_{all}. Tertile cut points were fixed on telecom development tasks.

age. A q8 RTN-W3 comparison of restored attention against byte-matched random attention is incomplete, and we report no result from it. The W4 restoration comparison below is a separate, pre-specified design.

Success equivalence. Not established (Table 1).

Supplementary perturbation analyses (exploratory). These analyses perturb the quantized weights directly instead of forking trajectories, and they enter no claim in the main text. Each derived checkpoint takes every decoder linear from the BF16 model, from Q, or from an interpolation between them; all other tensors come from the BF16 model, and the endpoints reproduce BF16 and Q bit for bit. *Restoration location versus random modules.* On the GPTQ-W4 and AWQ-W4 checkpoints, restoring the BF16 weights of one layer third, of attention only or of the MLP only is compared with restoring randomly chosen modules with the same number of BF16 parameter bytes. [result pending: closed-loop success of located vs. byte-matched random restoration, with intervals]. *Error dose.* The residual intervention $W_\alpha = W + \alpha(W_{\text{Q}} - W)$ scales the quantizer's own weight error, with $\alpha \in \{0, 0.25, 0.5, 1, 2\}$; it is not a bit width. [result pending: action agreement with BF16 and closed-loop success as functions of α]. *Low-bit failure modes.* [result pending: how W3 and lower-bit checkpoints fail in closed loop, by failure category].

M Provenance

Every number in the paper is traceable to released analysis outputs. Figures are drawn by a released plotting script from those outputs, with assertions on every value the text quotes, and the TikZ figures are typeset from numbers in the LaTeX source.

Cell (weight)	states	G	H	F	$G - H$	mean I	off-diag. %	exp. %	κ	rev. % (n)
q4 GPTQ-W4 (1/2)	385	$-3.1\,[-9.0, +2.8]$	$+0.5\,[-5.4, +6.4]$	$+1.8\,[-4.8, +8.4]$	$-3.6\,[-9.1, +1.8]$	$-4.4\,[-12.5, +3.7]$	19.7	27.0	0.269	3.4 (13)
q8 GPTQ-W4 (1/4)	399	$-5.8\,[-10.2, -1.3]$	$-4.0\,[-10.2, +2.2]$	$-5.3\,[-11.6, +1.1]$	$-1.8\,[-6.4, +2.9]$	$-4.5\,[-10.4, +1.4]$	19.5	23.1	0.155	3.0 (12)
q8 AWQ-W4 (1/4)	396	$-6.6\,[-13.0, -0.1]$	$-8.3\,[-15.3, -1.3]$	$-8.3\,[-15.5, -1.2]$	$+1.8\,[-3.3, +6.8]$	$-6.6\,[-14.5, +1.4]$	23.7	26.0	0.088	1.0 (4)
Weighted pool	1180	$-4.6\,[-7.8, -1.5]$	$-2.8\,[-6.3, +0.6]$	$-2.5\,[-6.2, +1.3]$	$-1.8\,[-4.8, +1.1]$	$-5.0\,[-9.6, -0.4]$	20.7	25.7	0.195	2.7

Table 18: Telecom holdout, *post-hoc* R2 estimands on divergent states (descriptive; not part of the pre-registered confirmation). pp; crossed task$\times$block 95% intervals; pool weights as pre-registered. Exp. and κ: marginal-preserving pairing agreement control; rev.: strict reversals.